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Actions required to make use of Exposure Resistance Classes for Concrete Structures in a contractor's business

Thesis in collaboration with NCC AB



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ABSTRACT

The concrete construction industry in Sweden, but also more broadly in Europe, is undergoing a transition in sustainable structural engineering practices regarding how concrete is designed at recipe level, particularly through changes in binder composition and the increased use of supplementary cementitious materials (SCMs). This shift is driven by the need to reduce CO_2 emissions and EU legislation, which affects the prices of cement. The new generation of Eurocode 2 introduces Exposure Resistance Classes (ERC), which represent a shift towards performance-based approach to durability design and concrete specification. This thesis investigates how Exposure Resistance Classes can be implemented in the Swedish concrete and construction industry, with particular focus on the implications for contractors and their workflow.

The study is based on a literature review combined with semi-structured interviews with experts in the field including contractors, structural engineers, industry experts, and concrete producers. The collected material was analyzed thematically to identify perceived benefits, barriers, changes in the industry, most important properties of the fresh concrete, and recommendations for implementation.

The findings show that ERC can support the transition of the industry towards more sustainable practices, through easier routes to use alternative binders in the concrete mixes at higher replacement levels than common. However, several barriers are present that slow down the implementation. These include limited industry knowledge, uncertainty regarding testing methods, limited laboratory capacity, and concerns regarding responsibility and liability. The results also show that changes in concrete composition may affect fresh properties, construction planning, and site workflows.

The thesis concludes that Exposure Resistance Classes have significant potential to support more sustainable concrete construction, but their successful implementation requires coordinated action between authorities, designers, contractors, concrete producers, and clients. Clear national rules, improved competence, reliable testing procedures, and early collaboration between actors are necessary for ERC to become practically applicable in Swedish construction projects.

Keywords: Eurocode 2, Exposure Resistance Classes, ERC, concrete durability, performance-based design, concrete structures, sustainable concrete.

SAMMANFATTNING

Betongbyggnadsindustrin i Sverige, men även mer generellt i Europa, genomgår en omställning mot mer hållbara konstruktionsmetoder. En viktig del av denna omställning berör hur betong utformas på receptnivå, särskilt genom förändringar i bindemedelssammansättning och ökad användning av supplementära cementbaserade material (SCM). Denna utveckling drivs av behovet att minska CO_2 utsläpp samt av EU lagstiftning, vilket påverkar cementpriserna. Den nya generationen av Eurokod 2 introducerar exponeringsresistansklasser (ERC), vilka representerar en övergång mot ett mer prestandabaserat angreppssätt för beständighetsdimensionering och betongspecifikation. Denna studie undersöker hur exponeringsresistansklasser kan implementeras i den svenska betong- och byggindustrin, med särskilt fokus på konsekvenserna för entreprenörer och deras arbetsflöden.

Studien baseras på en litteraturstudie kombinerad med semistrukturerade intervjuer med experter inom området, inklusive entreprenörer, konstruktörer, branschexperter och betongproducenter. Det insamlade materialet analyserades tematiskt för att identifiera upplevda fördelar, hinder, förändringar i branschen, de viktigaste egenskaperna hos färsk betong samt rekommendationer för implementering.

Resultaten visar att ERC kan stödja branschens omställning mot mer hållbara metoder genom att skapa enklare vägar för att använda alternativa bindemedel i betongblandningar vid högre ersättningsnivåer än vad som är vanligt idag. Samtidigt finns flera hinder som kan bromsa implementeringen. Dessa inkluderar begränsad kunskap inom branschen, osäkerhet kring provningsmetoder, begränsad laboratoriekapacitet samt frågor kopplade till ansvar och ansvarsfördelning. Resultaten visar även att förändringar i betongens sammansättning kan påverka färska egenskaper, produktionsplanering och arbetsflöden på byggarbetsplatsen.

Studien drar slutsatsen att exponeringsresistansklasser har betydande potential att stödja ett mer hållbart betongbyggande, men att en framgångsrik implementering kräver samordnade insatser mellan myndigheter, konstruktörer, entreprenörer, betongproducenter och beställare. Tydliga nationella regler, ökad kompetens, tillförlitliga provningsmetoder och tidig samverkan mellan aktörer är nödvändiga för att ERC ska bli praktiskt tillämpbart i svenska byggprojekt.

Nyckelord: Eurokod 2, exponeringsresistansklasser, ERC, betongbeständighet, prestandabaserad dimensionering, betongkonstruktioner, hållbar betong.

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Chapter 1

Introduction

1.1 Background and problem description

With the equivalent of three tonnes per year used for every person in the world, concrete is the second most consumed material in the world [Colin R. Gagg, 2014]. Cement is the key ingredient in the concrete mix, and its production emits approximately 8% of the global CO_2 emissions [Robbie M. Andrew, 2025].

Concrete construction in Sweden, but also in Europe, is currently undergoing a transition in sustainable structural engineering practices regarding how concrete is designed at recipe level, particularly through changes in binder composition and the increased use of supplementary cementitious materials (SCMs) [Leon Black, 2020]. This shift is driven by the need to reduce the CO_2 emissions associated with cement production, especially those related to clinker, which is the main source of cement-process emissions. In parallel, European climate-policy instruments such as the EU Emissions Trading System (EU ETS) and the Carbon Border Adjustment Mechanism (CBAM) are increasing the carbon cost associated with emission-intensive cement production and imports, thereby strengthening the incentive to reduce clinker content and adopt lower-carbon binder systems [Alain Mermier, 2023].

This transition affects both durability and production from a contractor's perspective, because changing the composition of the binder influences the properties of concrete, especially its fresh properties. This is therefore a key consequence for contractors that must be analyzed, assessed, and dealt with. This risk must be minimized and opportunities to deliver strong and resource-efficient projects, both economically and in terms of CO_2 emissions, must be identified.

The second generation of Eurocode 2 (prEN 1992) introduced the Exposure Resistance Classes (ERC). ERC are used to classify concrete with respect to resistance against corrosion induced by carbon (classes XRC) or chlorides (classes XRSD), and damage caused by thaw/freeze attack (XRF). They link the minimum cover of reinforcement required for durability $C_{\mathrm{min,dur}}$ to Design Service Life and Exposure Class. The purpose of the introduction of ERC is to reduce "unnecessary margins" for proven materials and faster/safer introduction of innovative materials [Marcus Peterson, 2023]. The concept of ERC aims to improve the connection between environmental exposure, properties of concrete, and technical service life of structures, by introducing a more performance-based and verifiable methodology for durability design. This requires new working methods and processes for the contractor in order to find the right concrete mix for

the unique conditions of each project regarding climate impact and buildability. ERC has not yet been fully implemented into Swedish mandatory regulations. This creates uncertainty for contractors, designers, and concrete producers regarding how ERC should be applied in practice.

1.2 Research gap and uncertainties

Through the studies shown in the Frame of Reference, it can be understood that there is a strong technical support concerning the use of alternative binders in the concrete mix. Research has shown that materials such as fly ash, GGBS, silica fume, and LC^3 can meet or exceed the durability performance of Portland cement [Hanzlah Akhlaq et al., 2025, Edward G. Moffatt et al., 2017, Yuvaraj Dhandapani et al., 2018]. In addition, testing methods exist to verify this performance. The limitations of the current prescriptive framework have been well-documented [R. Douglas Hooton, 2019, Mark Gavin Alexander, 2018], and the transition toward performance-based design is well-supported. However, several uncertainties remain that are relevant to the practical implementation of ERC by the actors in the Swedish construction industry.

One area of uncertainty is how the use of alternative binders affects the properties of fresh concrete and the construction workflow on site. As discussed in the Frame of Reference, fly ash replacement increases setting time and reduces early-age strength development [Björn Johannesson, 2012], and increasing silica fume content reduces workability [M. Mazloom et al., 2004]. These changes in material behavior may affect construction activities such as adding and removing the formwork and curing, which in turn can influence scheduling and planning. How contractors in Sweden are specifically affected by these changes has not been studied in detail (RQ3).

Another uncertainty relates to the testing required under the performance-based approach. Methods for assessing carbonation resistance, chloride ingress, and freeze-thaw resistance are available [Thomas A. Harrison et al., 2012, Jorge Pontes et al., 2023, Kayaalp and Yaliniz, 2025]. However, they typically require extended test durations, introducing financial and logistical challenges [Imrose B. Muhit et al., 2025]. Furthermore, the reliability of these methods has been criticized. Accelerated carbonation tests have been criticized for using CO_2 concentrations high enough to change the microstructure of concrete in ways that would not occur during natural exposure [Ahmed Merah, 2024]. Similarly, the theoretical model underlying the Rapid Chloride Migration Test has been described as oversimplified, which can lead to inaccurate estimations of the chloride migration coefficient [P. Spiesz and H. J. H Brouwers, 2010]. These limitations raise questions about whether current testing methods fully represent the long-term durability of concrete in real conditions (RQ2).

Finally, ERC has not been implemented in Swedish regulations yet. As reviewed in Section 2.6, there is currently no legally binding inclusion of ERC in Swedish building regulations, and businesses must still rely on prescriptive methods until national adaptations are published [SBUF, 2026a]. Ongoing SBUF projects are working toward generating the technical basis for national rules [SBUF, 2026b], but the timeline and form of implementation remain uncertain (RQ5).

A clear gap can therefore be identified in the existing literature regarding the practical, technical, and business-related implications of introducing ERC in the Swedish construction industry. This thesis aims to fill this gap by analyzing the stages of the design and delivery process, identifying the relevant stakeholders, and examining how they are affected by the shift from prescriptive to performance-based design. It further aims to understand which properties of the fresh concrete

are most influencing in the construction sites. Through the information received from the Frame of Reference and interviews with industry experts, the study seeks to identify both the potential benefits and the practical challenges of ERC implementation, and to determine what technical and organizational measures are needed to ensure quality, productivity, and buildability (RQ1–RQ5).

1.3 Aim and Purpose

The aim of this project is to identify and analyze the actions required for a contractor to effectively apply the Environmental Resistance Classes (ERC) in the design of concrete structures, in accordance with the second generation of Eurocode 2 (prEN 1992), EN 206 and national annexes. The thesis seeks to clarify how ERC can be translated from a regulatory concept into practical use within a contractor's business, considering technical and organizational aspects. In particular, the aim is to investigate how ERC affect engineering assessments related to concrete composition, durability, and properties, and how do these influences affect a contractor's business. The purpose of the thesis is to support contractors in using ERC as a methodological, practical, and strategic tool to enable the use of alternative, low-clinker and climate-efficient binder compositions while maintaining required durability and buildability.

Specific research questions intended to be answered are:

- **RQ1:** What are the potential advantages that ERC brings in concrete construction projects in Sweden?
- **RQ2:** What are the barriers, including uncertainties and risks, that currently limit the practical implementation of ERC in concrete construction projects in Sweden?
- **RQ3:** How do the properties of the fresh concrete change and how do they affect the workflow?
- **RQ4:** What technical and organizational actions are required within a contractor's working process to implement ERC, to efficiently translate from a regulatory concept to practical use?
- **RQ5:** What is the status of regulatory progress in Sweden and when will ERC be implemented?

By examining ERC in relation to existing working methods, the thesis aims to contribute to a clearer understanding of how contractors can adapt their processes to the evolving regulations. The results are intended to provide guidance that supports decision-making and systematic implementation of ERC within a contractor's development processes and structural engineering fieldwork.

1.4 Limitations

This study focuses on the investigation of ERC as a concept for the durability specification of concrete structures with respect to Swedish construction practice. The study is primarily conducted from a contractor's perspective, with reference to European standards and regulatory documents. However, this study does not include experimental laboratory testing, field measurements, or full-scale durability testing of concrete materials. Performance verification of ERC

is assessed based on existing standards, literature, and documented methodologies rather than own experimental data. Also, the thesis does not aim to develop specific concrete mix designs, nor to prescribe numerical models for binder composition, water-binder ratio, or cover depth.

Chapter 2

Frame of reference

This chapter establishes the frame of reference for the thesis by reviewing the main technical, regulatory, and practical aspects related to concrete durability design and the introduction of Exposure Resistance Classes (ERC). First, the chapter examines Portland cement, including its raw materials, production, hydration processes, admixtures, and possible alternative binders, in order to explain the material-related factors that govern concrete performance. It then discusses the limitations of the prescriptive approach in the first generation of Eurocode 2 and the motivation for the transition toward the performance-based approach introduced in the new Eurocode. In addition, the chapter presents relevant testing methods, previous studies, and the current status of ERC implementation in Sweden. Finally, the discussion is linked to the building design and construction process, where material selection, fresh concrete requirements, and stakeholder decisions become critical for practical implementation.

2.1 Portland cement

Unless otherwise stated, the content of this section is based on *Introduction to Concrete Technology: Lecture Notes* by Björn Johannesson [2012].

Portland cement is a hydraulic binder made from calcined limestone and clay, ground with gypsum. It is the main ingredient in concrete and mortar, providing strength and durability to structures. Different kinds of Portland cement are produced, e.g., ordinary Portland cement, sulfate-resistant Portland cement and rapid-hardening Portland cement.

2.1.1 Raw materials

The principal raw materials for the production of Portland cements are limestone, chalk, clay and sand. Limestone is the dominant calcareous component in the raw mix, and its primarily function is to supply calcium carbonate ($CaCO_3$), that will be converted into calcium oxide during burning. Often, the impure type of limestone, which is referred to as 'cement rock', is used. Besides $CaCO_3$, a cement rock contains silicic acid, aluminum oxides, and iron oxides. However, the cement rock should contain at least 80 weight percentage (wt%) $CaCO_3$, in order to use it without the addition of pure limestone.

Clay is the main source of silica, alumina, and iron-bearing components. A key process-relevant material property is that silica in clay is more reactive than quartz, meaning that clay can act as

a more favorable SiO_2 source than quartz sand in clinker formation reactions. Silica in clay has a much greater specific surface area and therefore is much more reactive at comparably lower temperatures than the one required in reaction. Sand is a silica corrective in the production flow.

Besides the principal raw materials, waste-derived constituents can be used in the production of Portland cement, to regulate the chemical composition of the raw material. Materials such as slag or fly ash are commonly used. However, these materials can be potentially problematic due to large variability in chemical composition, which can complicate the process control and the consistency of the quality of the cement.

2.1.2 Cement production

The cement production is a process that involves various steps. The process begins with crushing and grinding of the raw materials into a fine raw material powder. The approximate weight percentage of the raw materials is 70% limestone ($CaCO_3$), 20% quartz sand (SiO_2), and 10% clay, rich in alumina and iron oxides. It is important to grind the raw material into a powder before it is fed into the kiln in order to avoid occurrence of free lime in the finished product and to optimize the chemical reactions in the kiln. A raw material preparation is divided into a 'wet method' and a 'dry method'. In the wet method, when using $CaCo_3$ occurring as soft chalk, it can be mixed with clay and water to produce slurry. In this case, extra energy will be consumed in the evaporation of the water from the slurry in the cement kiln. Figure 2.1 shows the stages of cement production.

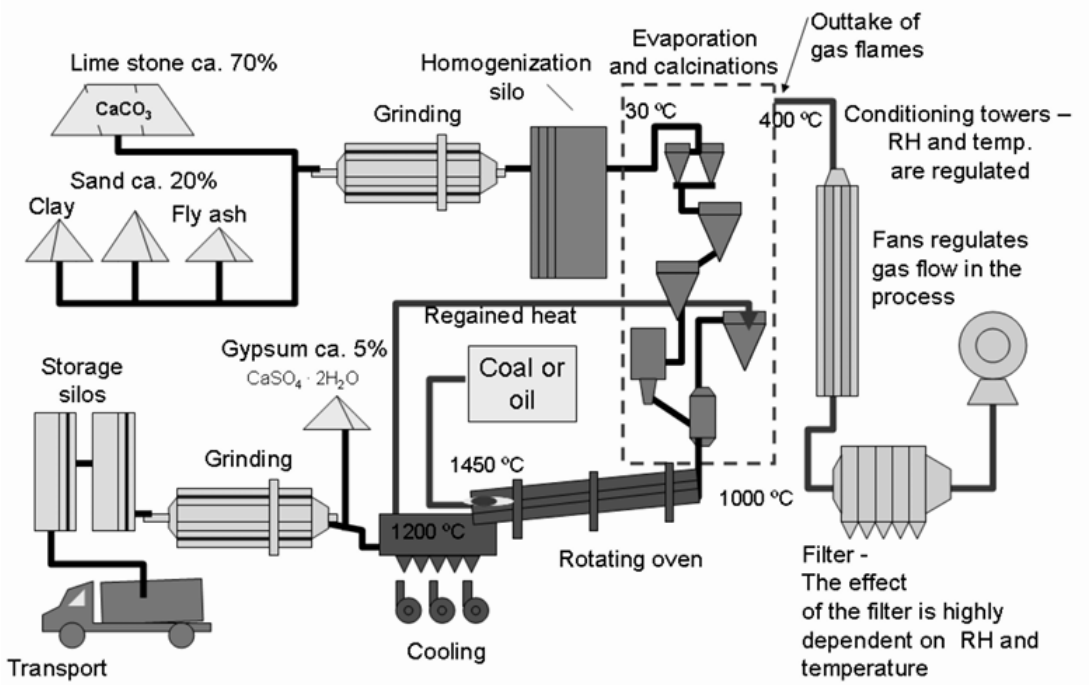


Figure 2.1: Schematic picture of cement production [Björn Johannesson, 2012]

The raw material composition is controlled by some key values, which the most common ones are called Silica Modulus (S.M.), Alumina Modulus (A.M.) and Lime Saturation Factor (L.S.F.). The Silica Modulus is used to check and adjust the material composition to the desired amount of melt in the cement kiln. It is defined as

$$\text{Silica Modulus (S.M.)} = \frac{\text{SiO}_2}{\text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3} \tag{2.1}$$

Optimal silica modules lie in the range of S.M. = 2.3 - 2.8, depending on the plant operation and the cement clinker quality. When the silica modulus increases the amount of the melt in the burning process decreases and vice versa. If the amount of melt in the burned material becomes too low, the formation of clinker nodules will not occur and the required chemical reactions will not be completed at normal maximum temperatures.

The Alumina Modulus is defined as

$$\text{Alumina Modulus (A.M.)} = \frac{\text{Al}_2\text{O}_3}{\text{Fe}_2\text{O}_3} \tag{2.2}$$

Optimal alumina modules lie in the range of A.M. = 1.6 - 2.0, which indicate the optimal operational issues and product quality. Approximately a value of A.M. = 1.6 gives the lowest melting temperature in which a melt is created. Alumina modules also affect the of formation of cement clinker nodules.

The Lime Saturation Factor is defined as

$$\text{Lime Saturation Factor (L.S.F.)} = \frac{\text{CaO}}{2.8\,\text{SiO}_2 + 1.18\,\text{Al}_2\text{O}_3 + 0.65\,\text{Fe}_2\text{O}_3} \tag{2.3}$$

Optimal values for the Lime Saturation Factor is in the range of L.S.F. = 0.92 - 0.96. L.S.F. is used to control that the reaction of free lime CaO in the kiln becomes complete, so there is no free lime in the produced cement. Based on trial and error methods, CaO reacts completely when L.S.F. < 1.

The cement burning process can be divided into four main stages: preheating and evaporation of free water, calcination, high-temperature burning, and cooling. Depending on the type of cement production method, the raw material is either fed directly into the calcining cyclones or in a preheating process, to evaporate free water and weakly bound water incorporated in the clay. This process is usually referred to as the decomposition of clay to produce meta-clay, that is,

$$\text{Clay} \rightarrow \text{Meta-clay} + \text{H}_2\text{O(g)} \tag{2.4}$$

A key reaction here is the calcination of limestone, $CaCO_3$, into CaO with the release of CO_2. This process occurs in temperatures ranging between 600-900 °C. The calcination process is

$$\text{CaCO}_3\text{(s)} \rightarrow \text{CaO(s)} + \text{CO}_2\text{(g)} \tag{2.5}$$

Simultaneously with calcination, meta-clay reacts slowly with free lime to form calcium combined with silica (CS), alumina (CA) and iron oxides (CF). Approximately 900 kg CO_2 per 1 ton of cement produced is emitted in the calcining process.

Continued heating of the materials in the kiln the CS phase will bind free lime in a solid-state type of reaction to form belite:

$$\mathrm{CaO \cdot SiO_2(s) + CaO(s) \rightarrow 2CaO \cdot SiO_2(s)} \tag{2.6}$$

After calcination, the material enters the top of a slightly tilted rotary kiln, with approximate measurements being roughly 50-250 m and about 5 m in diameter. This is the high-temperature burning stage of the process. The raw meal travels slowly toward the hot end, reaching a maximum temperature of about 1450 °C. A flame at the kiln outlet is produced by burning oil or powdered coal.

Only at these high temperatures will the belite be able to react with free lime to produce alite according to the equation

$$\mathrm{2CaO \cdot SiO_2(s) + CaO(s) + melt \rightarrow 3CaO \cdot SiO_2(s)} \tag{2.7}$$

Alite is the most important component in Portland cement and constitutes about 60 wt% of an ordinary Portland cement. At about 1400 °C, approximately 25-30 wt% of the kiln material is in a melt state (liquid state). The reaction described by (2.7) is slow and the residence time at maximum temperature is only a few minutes, so equilibrium is not reached and commercial cements retain some belite.

After leaving the kiln, the product is in the form of nodules called clinker. Clinker is partly melted and is slowly cooled to crystallize important phases. Cooling rate is critical for developing aluminate (C_3A) and ferrite (C_4AF) phases. Extremely fast cooling prevents formation of C_3A and C_4AF and instead yields a significant glass-like fraction. if clinker is cooled too fast, about 25% of the product can become a glass-like material that cannot react with water. Finally, clinker is ground to a fine powder, and about 5 wt% gypsum is added before grinding to regulate cement setting. Without gypsum, Portland cement would react too rapidly with water and produce a poor-strength material.

Carbon dioxide capture is a potential mitigation strategy for cement manufacturing emissions. One potential method is to capture CO_2 from the smokestacks of the cement production plants and lead the gas through a liquid in which carbon dioxide precipitation can occur in the form of sodium carbonate. A CO_2 capture route exists, which is based on producing an alkaline absorbent ($NaOH$) through electrolysis of sodium chloride ($NaCl$) in seawater and using this $NaOH$ to chemically bind CO_2 from flue gases. In this approach, electrolysis yields $NaOH$ and hydrochloric acid (HCl), after which the HCl is neutralized by reaction with magnesium-containing silicate rock, producing a sand-like residue and magnesium chloride ($MgCl_2$) . The CO_2-rich flue gas is then contacted with $NaOH$, forming carbonate and bicarbonate. CO_2 reacts with $NaOH$ to form sodium carbonate (Na_2CO_3) and water, and Na_2CO_3 can further react with CO_2 and water to form sodium bicarbonate ($NaHCO_3$). However, the described carbon dioxide approach described is energy-intensive and expensive.

2.1.3 Hydration of Portland cement and the influence of replacement materials

The hydration process of Portland cement is important to understanding the early behavior of concrete, since setting, hardening, and many practical casting-related issues depend directly on the chemical reactions that take place after water is added to cement. Portland cement is an anhydrous material mainly composed of the cement clinker materials, C_3S, C_2S, C_3A, and C_4AF. When mixed with water, these phases react and form new hydration products, and this overall process is referred to as hydration. In this context, setting describes the stiffening of the fresh paste without significant strength gain, while hardening refers to the following period during which the concrete develops compressive strength.

Among the clinker phases, C_3S is particularly important for early hydration and early strength development, while the reactions of the aluminate phase C_3A are highly significant for the setting behavior of cement. The hydration of Portland cement is therefore not governed by one single reaction, but by simultaneous interactions between the clinker phases and water. Also, nucleation, crystal growth, and hydrolysis are fundamental mechanisms in the hydration process. The hydroxide ions generated during these reactions play an important role in the chemical development of the cement paste and in the formation of the main hydration products.

A key aspect of ordinary Portland cement is the addition of gypsum, which is used to regulate setting. Without gypsum, the aluminate phase reacts too rapidly with water and may cause flash set, which is a phenomenon where the reaction happens too fast without gaining strength. The addition of gypsum therefore makes the cement usable in practice by controlling the early aluminate reactions. More generally, the hydration behavior of cement depends not only on the clinker composition, but also on factors such as fineness and the interactions between the different cement compounds.

When part of the Portland cement is replaced by pozzolanic materials, the hydration behaviour changes. Pozzolans are materials that react with calcium hydroxide and exhibit cementitious properties. Pozzolans such as fly ash, silica fume, calcined clay, and natural pozzolans, as well as slag are used in blended cements. In hydrated Portland cement systems, pozzolanic materials consume portlandite, $Ca(OH)_2$, and contribute to the formation of additional C-S-H. This conversion is beneficial because C-S-H has stronger bonding characteristics and forms a denser and less permeable structure than portlandite. As a result, the use of pozzolanic materials may improve the strength and permeability characteristics of the hardened material.

However, replacement materials also influence the early-age behavior of concrete, which is particularly relevant from a practical and contractor-oriented perspective. For fly ash-based concretes, the setting time is generally extended compared with ordinary Portland cement concrete. In addition, strength development at early ages, especially during the first 1-7 days, is lower because the pozzolanic reactions are relatively slow. Special care is required in cold weather concreting, since low temperatures retard fly ash reactions even more strongly than the ordinary Portland cement reactions. This makes high fly ash replacement particularly relevant when discussing practical risks related to casting, curing, stripping times, and early-site performance.

At the same time, fly ash may also improve certain fresh concrete properties. According to the book, fly ash often reduces bleeding, may lower water demand, and can improve workability due both to the increased paste volume and to the mostly spherical particle shape of the material. This means that replacement materials do not only introduce disadvantages; rather, they create

a balance between potential benefits in fresh concrete and possible disadvantages in terms of delayed setting and slower early hardening.

Practical limitations associated with high replacement levels of ordinary Portland cement are present. For fly ash, the amount of substitution depends on both the cement chemistry and the characteristics of the fly ash. It is stated that above about 25 wt% replacement there is usually not enough portlandite left for complete pozzolanic conversion, even though higher replacement levels are often used in practice. In such cases, part of the fly ash remains unreacted, although the reduction in strength may be moderated by a filler effect. The carbon content of fly ash is also important, since high carbon contents increase the water demand of the fresh concrete and may interfere with air-entraining admixtures. Furthermore, in high-strength concrete applications, the book recommends certain replacement ranges for mineral admixtures and notes that higher silica fume contents may lead to problems with plastic shrinkage.

2.1.4 Mineral and chemical admixtures in Portland cement

Admixtures are added to the concrete mix to improve the properties of the fresh and hardened concrete in a controlled way. They are divided into chemical admixtures and mineral admixtures. Chemical admixtures are typically surface-active agents, used to improve the fluidity of fresh concrete and/or to create an air-bubble system in the hardened material. Their practical functions include increasing workability without increasing the water content, retarding or accelerating the time of setting, reducing bleeding and segregation, improving pumpability, reducing the heat evolution at early hardening, and accelerating early-age strength gain. Mineral admixtures are waste materials, such as silica fume and fly ash. They are used for two main reasons, either to replace some of the cement used to produce concrete, or to improve the properties of fresh or hardened concrete.

For workability and rheology, the most important group is the water-reducing and set-controlling admixtures. These admixtures are used to produce concrete with relatively low water-cement ratios while still maintaining satisfactory workability and acceptable setting behavior. This is important because a lower water-cement ratio generally gives higher strength and lower permeability, and therefore often better durability. In addition, water-reducing admixtures help decrease segregation in fresh concrete, which contributes to a more homogeneous final product. Their basic mechanism is related to dispersion of cement particles, which reduces the amount of water trapped in particle agglomerates and allows more mixing water to contribute to flowability.

Based on chemical composition, five categories of water-reducing admixtures exist. *Lignosulfonate - based* admixtures provide effective water reduction and good strength development, but they may also entrain air and can sometimes produce sticky concrete that is difficult to place. *Hydroxylated polymer* admixtures are described as improving workability, reducing aggregate segregation, improving surface finish, and improving pumpability without entraining air. *Hydroxylated carboxylic acid-based* admixtures are less effective than other water-reducers and may require relatively high water contents, which in turn can lead to rapid bleeding. If this occurs, the bleeding water should be removed and the concrete should be re-vibrated to maintain homogeneity.

The most powerful water-reducing agents are the so-called superplasticizers, especially *sulfonated melamine* or *naphthalene formaldehyde condensates* and *polyether-polycarboxylates*. These high-range water-reducers can reduce the water content by up to about 30 vol% while still producing

highly flowable or almost self-leveling concrete. Their performance is related to their superior ability to disperse fine cement and pozzolan particles, therefore reducing the amount of water tied up in agglomerates. Some polyether-polycarboxylate admixtures retard early hydration but nevertheless develop strength faster than other admixtures, which is one reason why they are widely used in precast concrete and in applications where vibration work is to be minimized. At the same time, some high-range water-reducers provide high flowability only for about 30-60 minutes, after which the concrete behaves more like a conventional fresh concrete. This makes their time-dependent performance relevant for transport, casting logistics, and site execution.

Some of the main potential advantages of high-range water-reducers are:

1. Significant water reduction,
2. Reduced cement contents,
3. Increased workability,
4. Reduced effort required for placement,
5. More effective use of cement,
6. More rapid rate of early strength development,
7. Increasing long-term strength, and
8. Reduced permeability of the hardened concrete.

Some of the main potential disadvantages of the use of high-range water-reducers are:

1. Additional admixture cost (the concrete in-place cost may be reduced),
2. Slump loss greater than conventional concrete,
3. Modification of air-entraining admixture dosage,
4. Less responsive when used with some cement types,
5. Mild discoloration of light-colored types of concretes, and
6. Air voids and color blemishes on exposed and formed finishes.

Retarding admixtures are used to delay the early hydration. Their practical purpose is to keep the concrete workable for longer periods of time, which can be beneficial in ready-mix transport, large pours, and concreting in conditions where rapid stiffening would be problematic. Their importance lies in extending the handling time of fresh concrete without necessarily increasing the water content. This allows greater flexibility during transport, placing, and finishing operations, especially when concreting conditions are demanding. Retarding admixtures are either organic-based or non-organic retarders, and they have the best effectiveness when mixed into the fresh concrete at some delayed time, approximately 10 minutes.

Accelerating admixtures serve the opposite function to retarding admixtures, namely to shorten the time to setting and/or to increase early-age strength development. This type of admixture is therefore particularly relevant when early demoulding, rapid construction progress, repair work, or casting under cold conditions is important. When concrete is exposed to low temperatures in the early stage of hydration, the strength development can be very slow or in extreme cases even be stopped entirely. From a practical perspective, accelerating admixtures can help reduce delays

associated with slow hardening and can support production systems where early strength is critical. Together, retarders and accelerators show that admixtures are a key means of adjusting concrete behaviour to the demands of the construction process rather than relying only on the basic cement-water system.

Air-entraining admixtures create small and stable air bubbles during mixing, and these bubbles remain in the hardened concrete as pressure-relieving regions during freeze-thaw exposure. There is a difference between entrained air and entrapped air. Normal mixing only traps about 1-2 vol% air, which is insufficient to create a proper protective air-void system. As a general rule, about 5-6 vol% entrained air is needed to make concrete resistant to freezing and thawing. The air voids should be small, typically around 0.20-0.25 mm, and the spacing between bubbles should also be limited. In this way, air-entraining admixtures are directly linked to durability rather than only to fresh concrete workability.

The performance of air-entraining admixtures is sensitive to production and mix conditions. The required dosage can be affected by pozzolans and slag, by the presence of other admixtures, by slump level, by temperature, and by the condition and loading of the concrete mixer. For example, increasing temperature reduces the air content, while lower temperatures may increase it significantly, meaning that dosage adjustments are necessary when conditions change. Similarly, a very high slump may reduce air stability, and the mixer condition influences how much air is actually entrained. This is important from a practical viewpoint because air entrainment is not simply a matter of selecting a product, but it also requires control during mixing and production to ensure that the intended durability performance is actually achieved.

2.1.5 Alternative binders to Portland cement

As summarized by Hanzlah Akhlaq et al. [2025] ordinary Portland Cement is a major source of worldwide CO_2 emissions. This environmental burden is mainly due to the energy-intensive nature of cement manufacturing and the release of carbon dioxide during limestone calcination. For these reasons, alternative binders have become more attractive in decreasing the impact of concrete production in the environment. With the introduction of the new Eurocode 2, a new alternative to concrete production has been allowed. However, structural performance of concrete and workability are factors that these new binders should match the performance of cement to be considered as adequate alternatives. The following subsections show different binder replacement options and their advantages and disadvantages.

Ground Granulated Blast Furnace Slag (GGBS)

GGBS is a secondary material derived as a waste by-product from the iron-making process in blast furnaces. If not reused, the disposal of GGBS poses significant environmental challenges, including landfill use and potential soil and water contamination. By incorporating it in concrete, its environmental challenges can be addressed. GGBS improves concrete's flowability and pumpability by reducing both plastic viscosity and yield stress. Hanzlah Akhlaq et al. [2025] prepared 30 mix designs with GGBS replacements ranging from 0% to 90%, with a total of 810 samples prepared for compressive, splitting tensile and flexural strength. They concluded that the optimal GGBS replacement level was between 30% -50%, where the best balance between workability and strength development was achieved.

Fly Ash

Fly ash is a byproduct of coal combustion, and it is recognized as an eco-friendly material as its usage helps to lower the carbon footprint of the cement industry. Based on a systematic analysis of more than 30000 data points from 213 studies, Chao-Qun Lye et al. found that including fly ash as a cement component usually increases the carbonation rate of concrete compared to Portland cement concrete, whether designed for equal water/cement ratio or equal strength. They concluded that this tendency to carbonate, which causes the risk of corrosion, depends heavily on the mixture recipe, curing duration, and exposure conditions. To avoid these effects and ensure durability, the study suggests that concrete that contains fly ash, requires a greater cover, or higher specified strength that is typically stated by current standards. Additionally, they observed that the relative carbonation rate between fly ash and Portland cement concrete remains consistent across both accelerated laboratory tests and natural exposure conditions [Chao-Qun Lye et al.].

However, studies have shown that the presence of fly ash in concrete in marine conditions results in lower depth of penetration, compared to concrete specimens without fly ash. Edward G. Moffatt et al. [2017] conducted a study on the durability performance of concrete incorporating high volumes of fly ash in harsh marine environments. Specimens contained 56 or 58% fly ash and various types of normal density and lightweight aggregates. The study found that the depth of penetration was found to be in excess of 10 mm for the concrete without fly ash, whereas the presence of fly ash significantly decreased the depth of penetration to approximately 30 and 40 mm in specimens containing either normal density or lightweight aggregate.

Silica Fume

Silica fume is one of the most popular pozzolans, whose addition to concrete mixtures results in lower porosity, permeability, and bleeding. M. Mazloom et al. [2004] conducted a study to find the effect of silica fume on mechanical properties of high-strength concrete. The aim of the study was to investigate the effects of binder systems containing different levels of silica fume on fresh and mechanical properties of concrete. The concrete mixes had a constant water/binder ratio of 0.35 and constant total binder content. The percentages of silica fume that replaced cement were 0%, 6%, 10% and 15%. The results of the study indicated that the higher the percentage of silica fume, the less the workability of the concrete. Contrarily, the short-term mechanical properties such as 28-day compressive strength and secant modulus improved.

Limestone Calcinated Clay Cement (LC^3)

When using a large percentage of replacement of cement to other binders such as fly ash and slag, the issue of early age properties of the concrete has risen. This has happened due to the delayed setting and strength development in the concrete containing these binders. LC^3 has shown promise in this regard. Yuvaraj Dhandapani et al. [2018] conducted a study on concrete containing LC^3 as binder, and concluded that this concrete shows comparable strength development characteristics to ordinary Portland cement and superior strength development than concrete containing fly ash. The performance against gas permeability and capillary water absorption suggests that LC^3 can give better performance compared to Portland cement and comparable performance to fly ash.

Natural Pozzolans in Swedish Cement production

Although SCMs have shown capabilities to replace large portions of ordinary Portland cement, their availability in Europe is expected to decline. A scenario analysis conducted by Nina Chi Johansson et al. [2026] predicts a decrease of fly ash by more than 70% by 2035 compared to 2025. GGBS production is expected to decrease by 40-60% too. Therefore, the supply of SCMs will be insufficient to meet demand, and alternatives to sustain a low clinker-to-cement ratio will be needed.

Natural pozzolans are defined as either raw (volcanic material, limestone) or calcined natural materials (burnt shale, calcined kaolinite) with pozzolanic properties. Historically, they are among the oldest materials that have been used in combination with lime for construction purposes [Rosario García Giménez et al., 2015]. In Sweden, Bascement Plus is a cement blended with natural pozzolan, produced by Heidelberg Materials in Slite. Bascement Plus can be used is most common concrete applications [Heidelberg Materials Cement Sverige].

BETCRETE 3.0 is a Swedish project led by RISE (Research Institutes of Sweden) designed to accelerate the transition to climate-neutral concrete Sweden RISE - Research Institute of Sweden [2026]. The project investigated a wide selection of natural and industrial by-products to Portland cement, driven by the declination of traditional SCMs. Table 2.1 shows the types of materials that were tested, the companies that carried out the tests, and their findings, based on the report BETCRETE 3.0 RISE [2025].

Table 2.1: Alternative binders and industrial actors in Sweden

Company that conducted the tests	**Type of alternative binder**	**Material**	**Findings**
SCHWENK Sverige AB	Natural sedimentary pozzolan	Opoka (Lithuania)	High reactivity, good availability near factories, already on the market in Lithuania
Heidelberg Materials Cement Sverige AB	Volcanic pozzolan	Natural volcanic pozzolan	High amorphous content, good durability, verified and launched for the Swedish market
Strängbetong (Consolis)	Bio ash	Oat husk ash and rice husk ash	Usable as a pozzolanic supplementary material
Ragn-Sells	Residual ash from waste process (Ash2Salt)	Latent hydraulic properties	Circular solution, further verification ongoing
Boliden AB	Industrial pozzolans	Mining and metallurgical industrial by-products	Developed material for low-carbon cement and special applications
Swecem AB	GGBS	Ground granulated blast furnace slag	Further testing and implementation of GGBS

The results of the project indicate that alternative binders can significantly improve concrete durability, provided that correct mix design and curing are applied. Also, the new material standard SS 137004:2024 enables the use of natural pozzolans as additions to concrete, followed by the update of SS 137003 with amendment T2 in 2025, which enables the concrete manufacturers to use natural pozzolans as additions in their products.

Despite the promising results in BETCRETE 3.0, several important challenges remain to fully implement alternative binders in concrete production. Key challenges include uncertainties in material availability, incomplete standardization for some materials, necessity on tests for long-term performance, economic competitiveness, and the need for comprehensive life-cycle assessments to ensure true environmental benefits.

2.2 Eurocode 2

2.2.1 First generation: EN 1992-1-1:2004

In the first generation of Eurocode 2, EN 1992-1-1:2004, durability design follows a prescriptive methodology [European Committee for Standardization (CEN), 2004]. Rather than verifying the actual resistance of the concrete, the standard relies on a combination of strict material recipes and geometric minimum requirement to meet the requirements of stability of the structure.

The design process begins with the identification of environmental conditions, based on the chemical and physical conditions to which the structure is exposed, in addition to the mechanical actions. These environmental actions are categorized into specific Exposure Classes, which range from XO (no risk of corrosion) to aggressive environments such as XC (corrosion induced by carbonation), XD/XS (chlorides from de-icing or seawater), XF (Freeze/Thaw attack), and XA (chemical attack). These descriptive labels are the primary step for determining all design parameters.

To reduce these environmental risks, the standard requires compliance with the limiting values defined in EN 206-100 and the national application of standards for EN 206, such as SS 137003 in Sweden [European Committee for Standardization (CEN), 2021, SIS, 2021]. Durability is assumed by enforcing maximum water-to-cement ratios, minimum cement contents, minimum concrete strength classes. In addition to the material composition, geometric requirements exist, in the form of minimum concrete cover (C_{nom}) to protect the reinforcement. This is calculated as the sum of the minimum cover required for durability (C_{min}), and a safety margin for execution deviation (ΔC_{dev}):

$$C_{nom} = C_{min} + \Delta C_{dev} \tag{2.8}$$

The standard allows for limited optimization through Table 4.3N. for example, the Structural Class can be reduced (allowing for thinner cover) if the concrete strength exceeds specific thresholds (if $\geq$ C30/37 in XO). However, ensuring a service life of 100 years requires increasing the Structural Class by two levels.

2.2.2 Limitations of EN 1992-1-1:2004

While the old system (EN 1992-1-1:2004) provides a standard baseline for safety, it is fixed on certain material recipes, which creates a barrier to innovation. It does not account for the actual performance of the binder, making it difficult to utilize low-clinker, eco-friendly concrete mixes that might fail this recipe check but pass a performance check. K.H. Obla [2010] argues that rigid limits on mixture proportions, specifically minimum cement content and maximum water-cement ratios, often discourage the use of modern, low-carbon materials. Because these rules are based on the historical performance of Portland cement, they often do not consider alternative binder systems that could offer at least equal durability with a lower carbon footprint.

The transition to performance-based standards is largely driven by the need for sustainability. Unlike prescriptive standards, a performance specification outlines the functional requirements of the concrete rather than dictating a strict recipe. It sets requirements for fresh concrete for constructability and for dry concrete to stay strong and durable throughout its service life. The specifications should provide the necessary flexibility to the contractor and producer to provide a mixture that meets the performance criteria and avoid limitations or requirements on the ingredients or proportions of concrete [K.H. Obla, 2010].

Another factor to be taken into consideration is the over-designing of structures, in which more material than needed is used. Mark Alexander and Hans Beushausen [2019] state that many structures or parts of structures do not experience severe deterioration, in which case attention to good construction practices is needed. For example, the exposure category XO of EN 206-100 is described as concrete with reinforcement inside buildings with very low air humidity. This environment is not aggressive to the concrete and it is expected for the concrete to last virtually indefinitely. Such cases make up a significant portion of structures and overly sophisticated design approaches are not required here.

In addition, the transition from a Portland-cement-centric model is further supported by evidence that high-volume replacement of clinker with supplementary cementitious materials can enhance specific durability factors beyond the capabilities of traditional mixes. M.D.A Thomas et al. [2002] demonstrate that concrete containing high volumes of fly ash (up to 50-60%) significantly outperforms pure Portland cement in chloride-rich environments. Their data show that Portland cement allows deep chloride penetration over time. On the other hand, high volume fly ash mixes create a significantly more disconnected pore structure, effectively "locking out" chlorides in tidal and splash zones. This research highlights a critical flaw in prescriptive codes, where they focus on cement content as a representative for quality, older standards often restrict the use of high supplementary cementitious materials mixes that are technically superior for long-term service life in aggressive conditions.

2.2.3 The need for prEN1992 and future challenges

Consequently, the cumulative limitations of the prescriptive framework, specifically its inability to accommodate alternative binders and its tendency toward inefficient over-design, have rendered the first generation of Eurocode 2 insufficient to adapt to modern sustainable construction. R. Douglas Hooton [2019] provides expert validation that this is a global issue, not just in Europe. He argues that current prescriptive codes (like minimum cement content) act as "barriers to innovation" and fail to ensure true durability in severe environments. The introduction of the second generation (prEN 1992) answers this need by shifting to a performance-based methodology that allows the use of low-carbon materials previously excluded by rigid recipes. However, this evolution creates a significant operational challenge for the industry, transforming how concrete is specified and verified. This creates uncertainty for practical business, which leads to the need for this analysis, which aims to develop implementation strategies required for contractors to navigate this shift successfully without technical or financial risk.

2.2.4 The new Eurocode - prEN 1992

The 2023 revision of Eurocode 2 introduces a significant methodological shift in the durability design of concrete. In the new Eurocode 2, ERC are introduced, moving away from prescriptive concrete definitions toward a performance-based classification [European Committee For Standardization (CEN), 2021]. It is stated that ERC should be used to classify concrete with respect to resistance against corrosion induced by carbon (classes XRC) or by chlorides (classes XRSD) and damage caused by thaw/freeze attack (XRF). This structure decouples the environmental part (Exposure Class) from the material's resistance (ERC), allowing engineers to optimize the concrete cover (C_{nom}) based on the specific resistance characteristics of the concrete mix. This process integrates the use of EN 206, where concrete performance is verified either through limiting values and/or, for some ERC, by proving the performance in meeting the specifications of

relevant physical characteristics determined using standardized test methods.

Figure 2.2 shows Table 6.3 in prEN 1992, which gives the minimum cover for concrete structures exposed to carbonation.

Table 6.3 (NDP) — Minimum concrete cover $c_{min,dur}$ for carbon reinforcing steel — Carbonation

ERC	Exposure class (carbonation)							
	XC1		XC2		XC3		XC4	
	Design service life (years)							
	50	100	50	100	50	100	50	100
XRC 0,5	10	10	10	10	10	10	10	10
XRC 1	10	10	10	10	10	15	10	15
XRC 2	10	15	10	15	15	25	15	25
XRC 3	10	15	15	20	20	30	20	30
XRC 4	10	20	15	25	25	35	25	40
XRC 5	15	25	20	30	25	45	30	45
XRC 6	15	25	25	35	35	55	40	55
XRC 7	15	30	25	40	40	60	45	60

NOTE 1 XRC classes for resistance against corrosion induced by carbonation are derived from the carbonation depth [mm] (characteristic value 90 % fractile) assumed to be obtained after 50 years under reference conditions (400 ppm CO_2 in a constant 65 %-*RH* environment and at 20 °C). The designation value of XRC has the dimension of a carbonation rate [mm/$\sqrt{}$(years)].

NOTE 2 The recommended minimum concrete cover values $c_{min,dur}$ assume execution and curing according to EN 13670 with at least execution class 2 and curing class 2.

NOTE 3 The minimum covers can be increased by an additional safety element $\Delta c_{dur,\gamma}$ considering special requirements (e.g. more extreme environmental conditions).

Figure 2.2: Concrete cover for structures exposed to carbonation European Committee For Standardization (CEN)

Figure 2.3 shows Table 6.4 in prEN 1992, which gives the minimum cover for concrete structures exposed to chloride ingress.

Table 6.4 (NDP) — Minimum concrete cover $c_{min,dur}$ for carbon reinforcing steel — Chlorides

ERC	Exposure class (chlorides)											
	XS1		XS2		XS3		XD1		XD2		XD3	
	Design service life (years)						Design service life (years)					
	50	100	50	100	50	100	50	100	50	100	50	100
XRDS 0,5	20	20	20	30	30	40	20	20	20	30	30	40
XRDS 1	20	25	25	35	35	45	20	25	25	35	35	45
XRDS 1,5	25	30	30	40	40	50	25	30	30	40	40	50
XRDS 2	25	30	35	45	45	55	25	30	35	45	45	55
XRDS 3	30	35	40	50	55	65	30	35	40	50	55	65
XRDS 4	30	40	50	60	60	80	30	40	50	60	60	80
XRDS 5	35	45	60	70	70	—	35	45	60	70	70	—
XRDS 6	40	50	65	80	—	—	40	50	65	80	—	—
XRDS 8	45	55	75	—	—	—	45	55	75	—	—	—
XRDS 10	50	65	80	—	—	—	50	65	80	—	—	—

NOTE 1 XRDS classes for resistance against corrosion induced by chloride ingress are derived from the depth of chlorides penetration [mm] (characteristic value 90 % fractile), corresponding to a reference chlorides concentration (0,6 % by mass of binder (cement + type II additions)), assumed to be obtained after 50 years on a concrete exposed to one-sided penetration of reference seawater (30 g/l NaCl) at 20 °C. The designation value of XRDS has the dimension of a diffusion coefficient [10^{-13} m^2/s].

NOTE 2 The recommended minimum concrete cover values $c_{min,dur}$ assume execution and curing according to EN 13670 with at least execution class 2 and curing class 2.

NOTE 3 The minimum covers can be increased by an additional safety element $\Delta c_{dur,\gamma}$ considering special requirements (e. g. more extreme environmental conditions).

Figure 2.3: Concrete cover for structures exposed to chloride ingress European Committee For Standardization (CEN)

2.2.5 Corrosion propagation phase

Corrosion of the reinforcement starts after de-passivation of the steel, and it usually induces loss in the steel section and cracks in the cover parallel to the reinforcement followed by damaging [CEN/TC 250/SC 2, 2018]. Figure 2.4 illustrates the corrosion caused by carbonation and chloride attack.

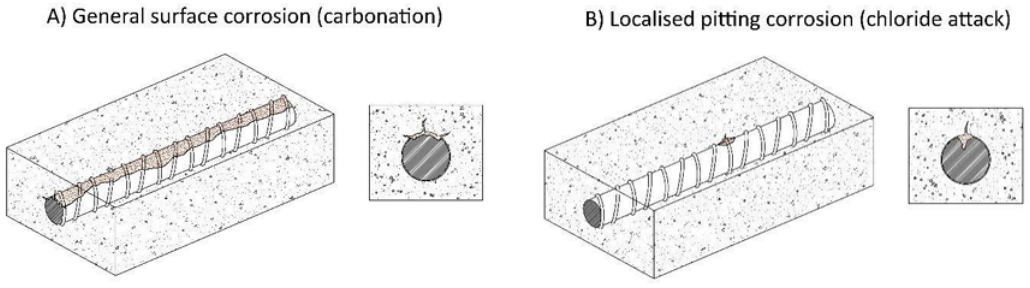


Figure 2.4: Corrosion by carbonation (A) and chloride attack (B). [CEN/TC 250/SC 2, 2018]

Before reaching these states, if the corrosion progresses very slowly, it may not produce any consequences neither in the external appearance of the structure nor in its performance. In present calculations, the propagation is considered only before any external sign is supposed to

be possibly detected.

2.2.6 Exposure Classes

The standard categorizes material resistance into three primary classes, addressing the dominant deterioration mechanisms for reinforced concrete [European Committee For Standardization (CEN), 2021].

Carbonation Resistance (Class XRC)

XRC classes are classes for resistance against corrosion induced by carbonation. They are derived from the carbonation depth (mm) assumed after 50 years under reference conditions. The class designation represents a carbonation rate with dimensions $mm/\sqrt{\text{years}}$. A lower XRC value means a slower rate of carbonation, therefore it allows for significantly reduced cover depths compared to higher values.

Chloride Resistance (Class XRDS)

XRDS are classes for resistance against corrosion induced by chloride ingress. These classes correspond to a reference chloride concentration at a specific depth after 50 years of exposure to seawater. The designation value of XRDS has the dimension of a diffusion coefficient ($10^{-13}m^2/s$). This parameter allows designers to specify concrete mixes with lower diffusion coefficients for marine (XS) or de-icing (XD) environments to reduce cover requirements.

Freeze/Thaw Attack

This exposure resistance class is not specified in the Eurocode. Instead, Eurocode refers to EN 206 for concrete to resist freeze/thaw attack [European Committee For Standardization (CEN), 2021]. With the new Eurocode 2, the method of choosing the most appropriate type of concrete to use in different environments moves from a prescriptive to a performance-based approach. The latter one, offers flexibility and encourages innovation, but often requires more testing.

The ERC can be satisfied by compliance with relevant limiting values for concrete composition and/or, for some ERC, by proving the performance in meeting the specification of relevant physical characteristics determined using standardized test methods. Adequate durability may be assumed against corrosion caused by carbonation or chloride ingress where cover to reinforcement is selected appropriate to the exposure class, exposure resistance class and the design service life and not less than the minimum cover for durability $c_{\text{min,dur}}$.

2.2.7 The workflow

The new Eurocode 2 decouples the linking between the Exposure Classes to concrete composition limits (water/cement ratio, minimum cement ratio) defined in EN 206 and National Annexes. These factors are decoupled by introducing a new performance classification, where the concrete's capacity to withstand environmental aggression is quantified by Exposure Resistance Classes (ERC), described in Subsection 2.2.4. The new ERC allow the designer to specify a requested resistance level based on the intended service life and cover depth, rather than specifying a concrete recipe. After the designer has chosen the Exposure Class, service life, and cover depth, the new Eurocode specifies which Exposure Resistance Class the concrete must specify. EN 206 is referenced by Eurocode 2 and offers two routes for compliance:

- The Descriptive Route, traditional tables that correlate the required ERC to composition limits.
- The Performance Route, verification through direct testing

2.3 Implementation of ERC in Sweden

Boverkets konstruktionsregler (EKS) are the current regulations that specify how the European construction standards (Eurocodes) are applied in Sweden. EKS incorporates Swedish national choices to account for local conditions such as climate, geology, and safety levels, called nationally selectable parameters. EKS together with the Eurocodes, form the mandatory codes for structural design. As of 1 July 2025, Sweden introduced a new consolidated set of building regulations that replace Boverket's Byggregler (BBR) and EKS as separate documents [Boverket, 2026b].

The concept of ERC is not currently part of the mandatory Swedish regulations in EKS. Efforts to integrate ERC into the Swedish system are ongoing. A SBUF project ,SBUF [2026b], is underway specifically to produce background material that informs how ERC can be introduced in Sweden. The project aims to generate the technical basis for national rules on ERC‘s application by comparing existing service-life design provisions with the new concepts of ERC in the new Eurocode 2.

According to another SBUF project, the Swedish implementation of ERC will occur through national choices published by SIS Technical Committees as part of national annexes to Eurocode 2 and updates to Swedish standards such as SS 137003. This implies that ERS's introduction will follow a voluntary national annex mechanism rather than an immediate change to statutory building regulations.

During this period, there is currently no formal, legally binding inclusion of the ERC methodology in Swedish EKS or the new building regulations, which means that businesses must still rely on prescriptive methods until adaptations are published SBUF [2026a].

2.4 Testing methods

The durability performance under ERC can be verified through standardized testing methods. Carbonation resistance is commonly assessed using accelerated carbonation tests (EN 12390-12), while chloride ingress resistance is frequently evaluated via migration or diffusion tests such as NT BUILD 492 and 443. Freeze-thaw resistance is typically determined using cyclic freezing procedures according to EN 12390-9. These tests enable the performance-based qualification of concrete mixtures.

2.4.1 Carbonation tests

To overcome the extended duration required for natural carbonation testing, carbonation resistance in performance-based durability design is commonly assessed through accelerated carbonation testing in which specimens are exposed to elevated CO_2 concentration and carbonation depth is measured using phenolphthalein. Thomas A. Harrison et al. [2012] conducted practical tests, from which they concluded that this method of testing successfully preserves the relative ranking of concrete mixes observed in natural exposure. They determined that a 10-week

exposure period gives the optimal balance of speed and reliability.

2.4.2 Chloride migration tests

The Rapid Chloride Migration Test (RCMT) is an accelerated laboratory method designed to assess the chloride penetration resistance of concrete [Jorge Pontes et al., 2023]. This method was developed by Luping Tang in 1996 and is standardized under NT Build 492. Its purpose is to determine the chloride migration coefficient ($D_{\mathrm{cl,RCMT}}$) in an expeditious manner, serving as a non-steady state method. Jorge Pontes et al. [2023] concluded in their study that RCMT classification successfully distinguishes the chloride resistance of different concretes under distinct exposure environments. RCMT results generally follow the same trends as long-term real exposure data regarding the influence of water/binder ratio.

2.4.3 Freeze/Thaw resistance

Freeze-thaw resistance is commonly evaluated through standardized cyclic freezing procedures, according to CEN/TS 12390-9, which simulate concrete exposure to freezing conditions in the presence of water or de-icing salts. In this method, concrete specimens are prepared with one exposed surface, while the other parts are sealed. After curing, specimens are subjected to repeated freeze-thaw cycles, with a saline solution in contact with the test surface to reproduce practical service conditions. Surface deterioration is periodically measured by collecting and weighing scaled material, allowing calculation of the mass loss per unit surface area. Comparative experimental studies using this method have demonstrated that it reliably differentiates between mixtures with good and poor freeze-thaw durability [Kayaalp and Yaliniz, 2025].

2.4.4 Computational modeling

Physical testing remains the standard for verification, however, its extended duration causes significant challenges when considering finances and logistics. Imrose B. Muhit et al. [2025] propose computational modeling as a complement to traditional methods, offering accelerated service life predictions for carbonation, chloride ingress, and freeze-thaw mechanisms. By integrating Artificial Intelligence and advanced algorithms, these models allow for more precise durability assessments without the extended durations of laboratory tests.

2.5 Previous studies on performance-based approach

In this subchapter, some previous studies are mentioned, to further understand research advancement on the ERC and to identify the uncertainties and the gap in current research.

The update of concrete design standards has been primarily driven by the limitations of previous code iterations that relied on prescriptive methods. R. Douglas Hooton [2019] argues that the "old" generation of Eurocode 2, which defined durability through deemed-to-satisfy rules like minimum cement content and maximum water/binder ratios, failed to take into account the real behavior of materials in severe environments. Mark Gavin Alexander [2018] critically evaluates traditional prescriptive specifications and has a similar argumentation to Hooton, and he adds that the deemed-to-satisfy parameters are unverifiable in the as-built structure. Alexander argues that prescriptive specifications come from an earlier era of lower material complexity and prevent innovation, particularly the use of alternative binders.

Empirical research on the economic impact of the EU Emissions Trading System (ETS) identifies a strong link between carbon legislation and rising product prices in the heavy industry sector. A comprehensive investigation of cost pass-through rates reveals relatively clear evidence that the cement industry has passed carbon costs onto customers [Johanna Cludius et al., 2020]. The International Energy Agency (IEA) confirms the cost-efficiency of this approach, basing its transition road-map on a "least-cost technology analysis" that prioritizes the reduction of the clinker-to-cement ratio. Furthermore, blended cements with a lower ratio of clinker to cement require less clinker and therefore generate less CO_2 emissions when manufactured, as the CO_2 footprint of some clinker substitutes is low or even zero [International Energy Agency, 2018].

Christoph Müller [2023] conducted an analysis on how standards support the sustainability of cement and concrete in Europe, describing how European standardization (CEN) is evolving to allow for more sustainable materials. He argues that standards are not just technical rules but strategic tools to enable the decarbonization of the industry. However, Müller concludes that the performance-based cures are not a "cure-all" for all projects. Instead, the paper recommends a hybrid method of work, where complex performance approaches are essential for critical infrastructure or severe conditions, and simple descriptive regulations should remain in place for low-risk applications to ensure speed and robustness. Müller further notes that in these standard cases, compressive strength can be used as a proxy for durability, provided that the relationship is verified.

The last statement by Müller can be supported by a proposal by Luisa María Gil-Martín et al. [2023], who derived a new function that adjusts the evolution of the carbonation front aligned with the Eurocode 2 values for carbonation-induced corrosion (XC exposure classes). The authors show that the evolution of the carbonation depth and the expected service life of a structure can be estimated using the new Eurocode 2, without relying on the resistance class against carbonation-induced corrosion. Instead, only the strength resistance of the concrete and the exposure class are used.

2.5.1 Criticism of accelerated durability tests

Ahmed Merah [2024] provides a comprehensive review of the various methods used for accelerated carbonation tests in concrete. The paper criticizes the accuracy of these tests by noting that high CO_2 concentrations can alter the microstructure and the strength class of concrete in ways that do not occur during natural exposure. It further explains that certain detection techniques, such as phenolphthalein spraying, fail to distinguish between loss of alkalinity from carbonation versus other acidic gases, potentially giving misleading results.

P. Spiesz and H. J. H Brouwers [2010] conclude in their study that the theoretical background in the Rapid Chloride Migration test is oversimplified and should be carefully treated . This model leads to an incorrect D_{RCM} (chloride-ion migration coefficient) estimation and the authors state that a new chloride transportation model should be developed to take into account the non-linear nature of chloride binding in concrete and non-equilibrium conditions between concentrations so that D_{RCM} could be calculated in an improved way.

In summary, the transition to Exposure Resistance Classes offers a new approach to include innovation and sustainability, yet the technical criticisms of current testing models suggest that a gap exists between theory and on-site application.

2.6 Current updates on the performance-based approach

Chris Clear and Tom A Harrison [2021] gave a presentation for the European Ready Mixed Concrete Organization (ERMCO) in March 2021. The presentation outlines the proposed framework for classifying concrete durability based on performance rather than prescriptive limits. In the presentation, ERC are introduced, followed by the integration with Standards (Eurocode 2 and EN 206). Clear and Harrison explain the working methodology, which is described in Subsection 2.2.7. The authors conclude that specifying durability by performance (ERC) and demonstrating it through testing will allow for more cost-effective and sustainable concrete solutions compared to prescriptive methods. However, they acknowledge that technical issues are present and need to be solved to produce a reliable, practical, and cost-efficient system.

Another presentation by Marcus Peterson [2023] of Svensk Betong at the Concrete Association ERC webinar, held on May 16th, 2023, focuses on the implementation of ERC in Sweden, their targeted users and benefits. He mentions that the vision goal for concrete in Sweden is to be climate neutral by 2045 and for climate-neutral concrete to be available on the market by 2030. The concrete for house construction shall reach half the climate impact in five years. In the presentation, he explains the ERC concept and their purpose. Peterson argues that, as an alternative to performing performance testing, it should also be possible to use tabular values for ERC based on modeling. In addition, he mentions that the need for national adaptation of ERC is high. The presentation outlines the roadmap for integrating ERC into European and Swedish standards, with the introduction of the next generation of Eurocode 2, published in 2023 with a 5-year transition period. A revision of EN 206 started in 2023, aiming to include ERC, and followed by the Swedish Standard (SS 137003) to include the Swedish application of ERC. In the presentation, a dedicated committee is mentioned to be needed within the Swedish Concrete Association to propose how ERC should be introduced, adapted for Sweden, and implemented to the SIS/TK 190. Then, SIS/TK 190 revises SS 137003 and also potentially influences the next EN 206 when it is out for consultation within CEN. Peterson mentions that the main actors in this shift are component material manufacturers, designers, concrete manufacturers, clients, managers, and researchers.

In a recent Swiss presentation by Stefanie Von Greve-Dierfeld and Christian Paglia [2025], the authors investigate the transition from prescriptive to performance-based durability design. The authors argue that the current Swiss framework (SIA 262), although semi-performance-based, still slows down decarbonization by enforcing rigid concrete types (named "Sortes") and a single, strict carbonation resistance $K_W = 6.5mm/year^{0.5}$ that many low-clinker binders cannot meet.

2.7 Different stages in the life of a building

Unless otherwise stated, the content of this section is based on the official website of the Swedish National Board of Housing, Building and Planning, [Boverket, 2026a].

2.7.1 Community Planning

Community planning is carried out by the municipality and concerns how land, water areas, buildings, and infrastructure are to be used, developed, and preserved. A comprehensive plan must be available in all municipalities, setting out the basic principles for how land and water should be used. Unlike the comprehensive plan, detailed plans are binding and apply to a specific

area. They regulate land use, building types, building rights, location, and design. If a building idea does not comply with the existing detailed plan, the client can apply to change or create a new one, though this process is typically time-consuming and costly.

2.7.2 Construction Idea

In the construction idea stage, the building's qualities and overall vision are established, and the relevant stakeholders are identified. Changes made at this early stage are relatively inexpensive compared to later phases.

Documents - Draft Proposals

Draft proposals include concept sketches produced to clarify and explore the project idea. Typically, the client hires an architect to prepare these sketches, which may also be used in the community planning stage if a detailed plan is being developed. During these early stages, the need to take ERC into account is minor or none.

After the documents are submitted to the Building Committee, a decision is made. If the proposal complies with the detailed plan, the project can proceed. If not, the design must be adjusted accordingly.

2.7.3 Design phase in the life of a building

The design phase begins with the client clarifying how the building will meet the project's requirements, needs, and wishes. These are specified through drawings, descriptions, and studies. For most buildings, the client is required to hire designers in areas such as fire safety, architecture, moisture safety, geotechnics, and acoustics. The design must account for site conditions including noise exposure, soil conditions, flood risks, and weather exposure.

The project idea is translated into documents of increasing detail as the design progresses, referred to as program documents, system documents, and construction documents.

2.7.4 Examples of documents

The documents produced ensure construction quality, verify legislative compliance, and satisfy civil law agreements between project stakeholders. At different stages, they increasingly contain more details.

Status: Program documents

Program documents are produced in the early design stages and provide the overall basis for decisions about continued design. They are not detailed enough to support a building permit application.

Status: System documents

System documents represent the stage where the client and designers evaluate which technical systems and materials best suit the project. All requirements, needs, and wishes are translated into drawings and descriptions. Building permits can be applied for based on system documents.

Status: Construction documents

Construction documents are the detailed documents that guide the contractor during the construction process. They present specific solutions for how all requirements, needs, and wishes are to be met.

The design phase is the critical link between the project's goals and physical reality. Material selection is one of the most impactful decisions made during this phase, particularly during the system documents stage. For concrete structures, this choice has become more complex with the introduction of the new Eurocode 2 (prEN 1992). The new standards change how concrete is calculated and specified, directly affecting construction documents and project participants. Understanding how these regulatory changes influence material choice is important for meeting safety requirements, workability, and sustainability goals.

2.7.5 The construction stage

During the construction stage, the building is constructed according to the documentation, with checks, inspections, and follow-ups conducted to ensure that all requirements are met. The contractor takes primary responsibility for executing the work, and site engineers and workers must have sufficient knowledge of the requirements placed on the building to ensure it is built correctly. Workplace visits by the Building Committee are conducted during construction to assess whether the inspection plan is followed. When the building is nearly complete, the client must notify the Building Committee so that a final consultation can be arranged. The building cannot be occupied until the Building Committee has issued a final decision.

Figure 2.5 shows a summary of the most important stages when planning, designing, and constructing a building. The shift from the building permit request and decision to the start notice marks the shift of the phases from designing to constructing.

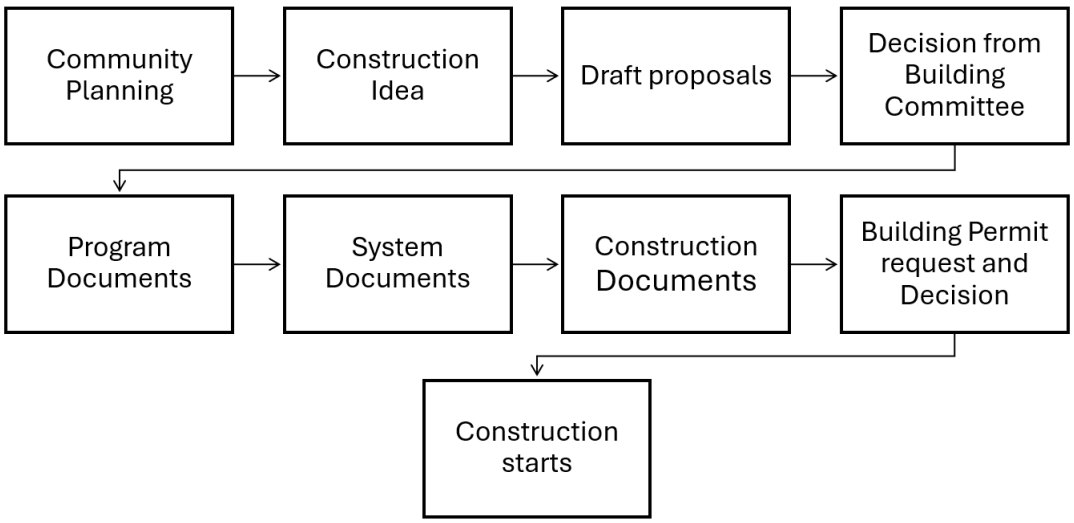


Figure 2.5: Stages of planning

2.7.6 Actors during each stage

Table 2.2 shows the main phases of the building process and the corresponding actors. *Kontrollansvarig* is a certified, independent role required under the Planning and Building Act,

responsible for coordinating and documenting that the legally required controls are carried out, while the client retains full legal responsibility.

Table 2.2: Main phases of the building process and associated actors

Phase	**Responsible**	**Participating actors**
Community Planning	Municipality	Client
Construction Idea	Client	Designers
Draft proposals	Client	Architects, Designers
Authority decision	Building Committee	Client
Program documents	Client	Designer, Expert
System & construction documents	Client	Designer, Expert
Building permit	Building Committee	Client
Planning for construction	Contractor	Client
Execution of construction	Contractor	Client
Workplace visit	Building Committee	Client, Kontrollansvarig
Final consultation	Building Committee	Client, Kontrollansvarig

2.8 Material selection

Concluding from the information provided by Boverket, material choices are made at an early stage of the project, specifically during the System Documents stage, so then the drawings and needed documents can be used for the building permit application.

For concrete structures, concrete selection is a complex decision-making process due to the need to balance the trade-offs created by conflicting requirements such as strength, sustainability, and resource consumption [Asad Ur Rehman Bajwa et al., 2025]. In addition, the characteristics of concrete depend on its targeted application and the required performance, demanded by the client Rawaz Kurda et al. [2019]. These arguments further underline that concrete mix decisions must occur during the design and documentation phases, when project requirements are translated into technical specifications and when it is easier and cheaper to make changes.

To handle such trade-offs, Multi-Criteria Decision Making (MCDM) methods exist. For example, TOPSIS (Technique for Order of Preference by Similarity to Ideal Solution) is a method that can be used to select the best binding material alternatives by integrating an objective. Asad Ur Rehman Bajwa et al. [2025] also argue that applying the MCDM methods is critical to the project's success and sustainability outcomes, and that these methods are most effective when used to facilitate the early conceptual design and product development process.

Similarly, *CONCRETop* is another multi-criteria decision method created by Rawaz Kurda et al. [2019], which can be used to optimize conventional or non-conventional concrete mixes. It uses three perspectives, quality performance, environmental impacts, and efficiency, and to identify the main differences between different mixes.

In addition to structural and sustainability considerations, with the ERC, there are also durability-related performance requirements based on the environmental conditions to which a structure is exposed. Properties of concrete, such as strength class, binder composition, water/binder ratio, and cover requirements are influenced in this stage.

2.9 Properties of the fresh concrete

Unless otherwise stated, the content of this section is based on the concrete book, *Betonghanbok Material. Del 1* by AB Svensk Byggtjänst [2017].

Fresh concrete is the phase of concrete from the time its component materials are mixed until it reaches a setting time that it can no longer be compacted. The properties of fresh concrete have a major impact on the properties of hardened concrete. Therefore, concrete must be formulated with both the cured required properties, such as strength, density, and resistance, but also the properties, such as workability, rheology, stability, and air content. When concrete has cured, it should be compact and homogeneous. To achieve these requirements, when concrete is on its fresh state, it must be workable and stable. Good workability enables fast casting and a good working environment, and it meets the requirements of the finished structure.

2.9.1 Requirements for fresh concrete

Regarding the requirements for fresh concrete, Eurocode 2 refers to the requirements in EN 206 to obtain good quality concrete structures. EN 206 specifies different types of property classes for concrete. For fresh concrete, these consist of consistency classes which are divided into classes for slump, compaction ration, spreadability, and flow slump. Basic requirements for concrete composition according to EN 206 are that the composition and component materials of the concrete shall be selected to meet the specified requirements for fresh and hardened concrete. EN 206 refers to standardized testing methods for fresh concrete, which are mainly based on the EN 12350 series. These testing methods test consistency, viscosity, permeability, separation resistance, air content, fiber content, and distribution.

The client usually sets requirements for strength, durability, density, and appearance, which are requirements on the hardened concrete. It is the designer's and the contractor's job to translate these requirements into requirements for the fresh concrete. The designer sets requirements related to strength and durability, which affect properties such as weight, air content, stone size, and cement type. The contractor sets requirements that are related to the execution on the construction site. These requirements include: workability, solidification time, temperature, pumpability, and stability. From the contractor's business perspective, these fresh properties are critical for optimizing labor and equipment use. The requirements of the concrete and how they are met should be discussed between the contractor and concrete manufacturer.

Knowledge of the properties of concrete at an early age is important to ensure rapid construction rate. This includes requirements such as early formwork removal and avoidance of cracks caused by moisture and temperature-induced movements. The concrete is considered to be at an "early age" when its properties undergo rapid development, which occurs until approximately 50% of the cement has hydrated, which usually occurs within 3 days. Figure 2.6 shows the stages of concrete's hardening process. During the first stage, the concrete is designed as fresh, a certain stiffening takes place, but the concrete is still workable. It is during this initial phase, the contractor depends on a predictable early strength gain to facilitate the finishing of the surface.

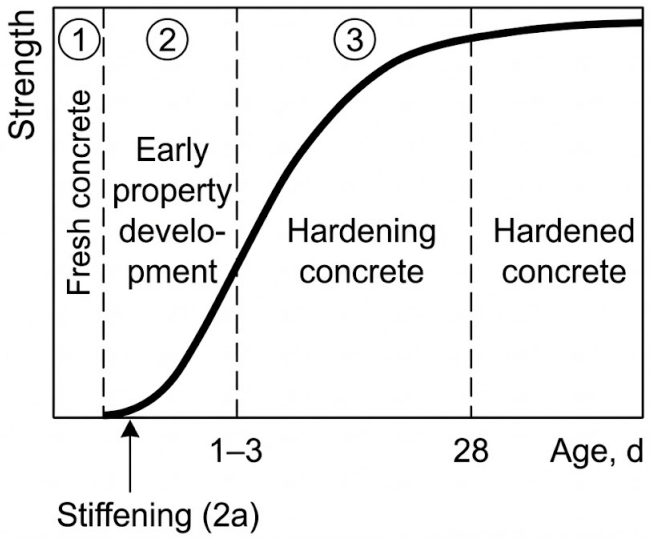


Figure 2.6: Schematic picture of stages of concrete's hardening process

During the second stage, the strength begins to increase, and the concrete is very sensitive to any type of loading. Internal drying occurs due to chemical reactions, causing autogenous shrinkage. During the third stage the actual hardening begins and concrete is also sensitive to drying, temperature effects, and the effect of external loads. By the end of this phase the concrete is almost fully not affected by external conditions and functions very similarly to the hardened concrete.

Knowledge of compressive strength at an early age is of great importance, to ensure time for formwork removal, avoiding early freezing and early cracking. The water-to-cement ratio strongly influences the strength development. Temperature also has a significant effect on strength development. Additionally, the contractor must manage the heat development and the resulting crack risk, as the exothermic nature of cement hydration can cause the inside of the concrete to be hotter than the outside, creating crack risk that can weaken the structure.

Plastic shrinkage occurs when concrete is drying and can cause cracking, which are called plastic shrinkage cracks. Both shrinkage and bleeding involve a de-watering of the concrete, resulting in consolidation of the material. Cracks normally appear as soon as the concrete surface becomes dry. The shrinkage rate depends mainly on the evaporation rate, which is affected by atmospheric conditions and by the concrete's temperature and exposure. Other factors are w/c ratio, cement type, dose of superplasticizer etc. The methods used to prevent cracking are water spraying, covering with plastic foil, water mist and covering, and application of membrane curing liquid. Effective management of early shrinkage is a key responsibility for the contractor to avoid surface defects that could lead to future maintenance liabilities.

Early autogenous shrinkage is, alongside plastic shrinkage, another cause of early crack formation, especially for modern concrete mixes with low w/c ratio (0.50-0.55). Because these concrete mixes have so little water, the chemical reactions inside consume the internal moisture, causing the shrinkage. The autogenous shrinkage occurs within the first 24 hours after pouring the concrete. Finally, the contractor must monitor the drying process prior to flooring, as residual moisture can cause adhesion failure or damage to floor coverings.

Workability describes the ease with which the concrete is poured. It is usually described with simplified methods such as slump, deformation coefficient and spreadability. Insufficient workability can result in reduced strength, reduced density or damaged surfaces.

Chapter 3

Methods

This study was conducted using qualitative research methodology, with a strong emphasis on regulatory analysis and industry application. A qualitative approach was considered appropriate because the study aims to explore perceptions, implementation needs, shifts in responsibilities, and concrete properties that are related to the introduction of ERC in Sweden. Since the topic is still developing and involves professional judgments across several actor groups, semi-structured interviews were selected as the main method for collecting empirical data. The qualitative interview data was complemented by an in-depth Frame of Reference, which provides the theoretical and technical background of the study and presents the current state of the topic in Sweden. This theoretical and technical background is used to formulate the questions for the interviews and to interpret the perspectives of the interviewed experts.

The Frame of Reference establishes the technical, regulatory, and practical background of the study. It examines Portland cement, its hydration process, admixtures, alternative binders, and the concrete properties relevant to both durability and construction practice. Both the previous and the newer generations of Eurocode 2 are analyzed, to establish the limitations of the prescriptive approach in the first generation of Eurocode 2 and the performance-based approach introduced in the newer generation. Associated testing methods, current implementation developments in Sweden, and previous research on the opportunities and uncertainties related to this shift are also described. The main stages of the building process are described, and the main actors in each phase are identified. These descriptions, together with the discussion of concrete properties, helped identify which properties are most critical at different stages of the building process and how they may influence actors' decisions.

3.1 Interviews

The main empirical data collection method used in this study was semi-structured interviews. In total, 11 interviews were conducted. The number of interviews was considered sufficient because the selected participants represented the main actor groups relevant to the study or had close professional interaction with them. Within the scope of this study, the final interviews produced limited new codes or perspectives, suggesting that sufficient thematic coverage had been reached for the purposes of the thesis. This was observed progressively within each group rather than only at the end of the full data collection process. The interviewees included structural designers, material science experts, professionals working with business development, concrete

suppliers, and experts involved in the regulatory implementation of ERC. These participants were selected through purposive sampling in order to capture the main perspectives relevant to the implementation of ERC in Sweden, including contractor-side decision-making, regulatory and technical development, and concrete suppliers.

3.1.1 Interview groups

The interviewees were divided into three generalized groups, namely:

- ***Contractor group***, consisting of structural designers, material science experts, and professionals working with business development. Their backgrounds included concrete production, civil engineering, building materials, and building physics. This group consisted of four participants.
- ***Industry experts group***, consisting of experts currently involved in the implementation of ERC in Sweden. This group consisted of three participants.
- ***Concrete suppliers group***, consisting of representatives from different concrete supplying companies. These participants worked either in technical roles or in sales-related roles within their companies. This group consisted of four participants.

These groups were selected in order to capture the main perspectives relevant to the research questions. The contractor group contributed primarily with perspectives related to practical implementation, business implications, and the barriers that contractors may face when adopting ERC. Since these participants are also closely connected to project clients and decision-making processes, they were able to provide informed perspectives on how ERC may be perceived from the client side as well. For this reason, and also considering that ERC is still at an early stage of implementation and that many clients currently have limited awareness of it, it was decided not to include clients as part of the interview process.

The industry experts group was included to provide insight into the current status of ERC implementation in Sweden. These participants were involved in different aspects of the implementation process, such as testing methods, calculation models, and regulatory development. Their interviews were used to identify the current progress of implementation, the barriers that remain, and the ways in which future regulations may support practical application.

The concrete suppliers group was included to examine how concrete suppliers currently understand ERC and how they perceive its practical implications. These interviews also contributed to the analysis of which concrete properties may be most affected by the transition toward a performance-based approach. Representatives from different concrete supplying companies located in different parts of Sweden were chosen to be interviewed, in order to understand whether geographical location influenced their perspectives.

3.1.2 The interview process

The interview process began by identifying relevant participants within the three groups. The first contact with the participants was conducted by the supervisors at NCC, after which the interviews were scheduled by the author of the thesis. Participation was voluntary, and all interviewees were informed that they could decline to answer any question they did not consider themselves sufficiently informed to answer.

The list of questions of the first group was tailored to each interviewee, as they had different profiles, backgrounds, and are involved in different fields within the company. In contrast, the same interview guide was used for all participants in the industry experts group and in the concrete suppliers group. The complete list of interview questions is presented in *Appendix 1.*

All interviews were conducted online via Microsoft Teams. The interviews were recorded and later transcribed with the support of an Artificial Intelligence (AI) transcription tool. The tool was selected specifically because it operates under a no-data-retention policy, meaning that interview recordings and transcriptions were not stored or used for training purposes by the tool provider. This choice was made to comply with data privacy requirements and to protect the confidentiality of participants. All participants were informed before and during the interview about the recording and transcription procedure, and their consent was obtained. The interviews lasted between 40 and 70 minutes, depending on the participant's role and level of involvement in the topic. All transcripts were manually reviewed in order to correct possible AI-generated errors and ensure the accuracy of the empirical material. During the manual review, particular attention was paid to technical terminology, which is where AI transcription tools are most likely to produce inaccurate outputs.

3.1.3 Ethical considerations

The author conducted this study under the supervision of NCC, a contractor company operating in the Swedish construction sector. The NCC supervisors facilitated the initial contact with participants across all three groups, which supported access to relevant professionals. Although NCC facilitated access to participants, the interview design and analysis were carried out by the author. The same thematic procedure was applied across all participant groups.

The study was conducted in accordance with the principles of informed consent, voluntary participation, confidentiality, and data protection. All participation was voluntary, and participants were informed of the purpose of the study prior to the interviews. Verbal consent to record and transcribe the interviews was obtained from all participants at the start of each session. To protect the anonymity of the interviewees, no participant is identified by name in this thesis. Quotes and statements are attributed only to the relevant group (e.g., Contractor group, Industry experts group, Concrete suppliers group) and, if relevant, to the professional role of the participant. Interview recordings were stored securely and accessed only by the author. Transcripts were anonymized before being used in the analysis.

3.2 Analysis of empirical data

The data collected from the interviews was analyzed using thematic analysis, based on the approach proposed by Virginia Braun and Victoria Clarke [2006], adapted to the structure of this study. Thematic analysis is a method for identifying, analyzing and reporting patterns within data. It minimally organizes and describes the data set in detail. The process of the analysis consists of six stages: familiarization, coding, clustering, refinement, synthesis, and interpretation.

Phase 1: Familiarization with the data. All transcripts were read multiple times until fully familiarized with the depth of the data collected. During this phase, preliminary ideas for coding were marked.

Phase 2: Generating initial codes. Codes identify a feature of the data that is considered interesting to the topic. During this phase, codes were created by coloring the statements of the interviewees, to capture explicit and implicit references to implementation of ERC and the properties of concrete affected.

Phase 3: Clustering and theme development. Codes were clustered into preliminary themes. During this phase, the analysis was shifted from individual codes to broader patterns across the data. The codes are grouped considering how different codes may combine to form an overarching theme. An Excel sheet was used to systematically store themes, sub-themes and the references supporting them. The full Excel sheet is provided in *Appendix 2*.

Phase 4-5: Refinement and definition. The clusters were iteratively refined to ensure internal consistency and empirical grounding. The final framework comprised six interdependent themes. The themes were selected to address the research questions, based on the data available from the interviews.

Phase 6: Synthesis and interpretation. In this phase, the qualitative insights were synthesized into a process model that explains how ERC will influence the construction industry in Sweden regarding concrete structures.

3.2.1 Trustworthiness

Several measures were taken to strengthen the trustworthiness of the analysis. To enhance credibility, the coding and theme development process was discussed with the supervisors at NCC throughout the analysis, providing a form of peer debriefing. All themes, codes, and supporting references were systematically documented in an Excel sheet (see *Appendix 2*), making the analytical process transparent and allowing the reasoning behind each theme to be traced back to the empirical data. The detailed descriptions of the participant groups, interview procedure, and research context provided in this chapter allow the reader to assess the relevance of the findings to other settings. Within the industry experts and concrete suppliers groups, the use of a consistent interview guide across all participants also ensured that the data was collected in a comparable way, supporting the dependability of the analysis.

3.2.2 Limitations

Several limitations of this study should be acknowledged. First, the number of interviewees was limited. Although the participants were selected to represent relevant perspectives from the concrete construction sector, the findings are not intended to be representative of the entire Swedish construction industry. Instead, the results should be understood as qualitative insights into current perceptions, challenges, and implementation needs related to ERC. All interviews were conducted online via Microsoft Teams, which, while practical given the geographical spread of participants across Sweden, may have limited the depth of interaction compared to in-person interviews.

The study is also limited in terms of actor representation. Although the interviewed participants represent several relevant groups, certain perspectives from the full project chain are absent. Clients were excluded from the interview process, as explained earlier in this chapter, given their currently limited awareness of ERC. However, the perspectives of on-site workers and site engineers, who are responsible for the practical execution of concrete work, are not represented. Their experiences of day-to-day implementation challenges could have added a valuable

operational dimension to the findings. Additionally, the perspective of regulatory authorities is only partially captured, as none of the interviewees represented such bodies directly.

Another limitation is that the level of prior knowledge about ERC varied among the interviewees. Some participants were already familiar with the concept, while others had only limited knowledge before the interview. As a result, some responses may reflect preliminary interpretations rather than experience from direct application in projects. At the same time, this variation provides useful insight into the current level of industry readiness and awareness.

As with all qualitative research, the analysis involves interpretation by the researcher. Although the coding process was carried out systematically and discussed with supervisors, the identification of codes and themes is partly dependent on the researcher's judgment. Different researchers may therefore have structured or interpreted parts of the data differently.

Finally, the study is mainly situated in the Swedish construction context. The findings are therefore influenced by Swedish regulations, industry structure, and ongoing standardization work. The results may not be directly transferable to countries where ERC implementation follows a different regulatory or market development.

Chapter 4

Results and Discussions

This chapter presents and discusses the main findings from the interview analysis. The findings are structured into six themes: benefits of ERC, barriers and uncertainties, changes in the industry, requirements for implementation, properties of the concrete affected, and status of development and regulatory progress. These themes were developed through the thematic analysis of the interview data and are discussed in relation to the Frame of Reference.

Before presenting the individual themes in detail, Table 4.1 provides an overview of the themes, codes, supporting empirical arguments, and example data sources. The table is intended to summarize the structure of the findings, while the following sections provide the detailed discussion, quotations, and interpretation.

Table 4.1: Overview of themes, codes, and supporting empirical arguments

Theme	Code	Main supporting argument from interviews	Data source
Benefits	Reduced climate impact	The possibility of using concrete mixes with lower cement content and higher proportions of alternative binders can help reduce the climate impact of concrete.	Contractor 0
	Support for environmental targets	A performance-based approach may support environmental targets by allowing concrete to be specified according to required properties rather than fixed composition.	Contractor 2; Contractor 3
	Freedom to use low-carbon concrete	ERC may provide greater freedom in mix design, allowing alternative binders to be used when performance can be verified.	Contractor 0; Concrete Supplier 2
	Easier introduction of new binders	Producers can test new mixes that contain alternative binders that are currently not used in the industry.	Industry Expert 2; Contractor 1
	Optimization	Concrete mixes can be optimized by using available materials more efficiently while still meeting durability requirements.	Industry Expert 2; Contractor 1
	Economic benefits	Higher allowable water/cement ratios and reduced cement contents may lower material costs and project costs.	Contractor 0
		From the supplier perspective, additional testing, documentation, or product development may increase concrete prices.	Industry Expert 0
	Projects or Exposure Classes with greater benefit	Large-scale projects and structures exposed to chlorides may benefit most from ERC because optimization potential is higher.	Contractor 3; Industry Expert 0
Barriers and uncertainties	Lack of knowledge	Structural engineers and other actors do not yet have sufficient knowledge about ERC and concrete properties in general.	Contractor 0; Industry Expert 1
		Smaller concrete suppliers may have less internal expertise than larger suppliers.	Industry Expert 1
	Struggle for smaller companies to comply	Smaller companies may face difficulties complying with ERC due to limited technical resources, expertise, laboratory capacity, and financial flexibility.	Industry Expert 0; Contractor 0
	Increased complexity	The use of a wider range of products combined with the lack of knowledge may increase complexity in specification and production..	Contractor 0; Concrete Supplier 2
		Economic complexity may arise if demand for new performance-based concrete products is uncertain.	Contractor 2

Continued on next page

Theme	Code	Main supporting argument from interviews	Data source
	Long transition time	Practical implementation of ERC is expected to require a long transition period.	Industry Expert 0; Concrete Supplier 2
	Client skepticism	Clients are often hesitant to try new mixes of concrete because they are familiar and comfortable with the old ones. Some clients do not believe in the presence of SCMs in the concrete mix.	Contractor 2; Contractor 3
Changes in the industry	Increased competitiveness	ERC may increase the competitiveness between concrete suppliers/contractors.	Contractor 2; Concrete Supplier 1
		ERC may create opportunities for new specialized suppliers to enter the market.	Contractor 2
	Shift in responsibilities	Structural engineers will still have an important role in prescribing, while concrete suppliers may take greater responsibility for mix development, testing, and documentation.	Industry Expert 1; Contractor 0
		A new type of prescriber with stronger knowledge of concrete properties may be needed.	Concrete Supplier 2
	Liability if performance fails	Liability remains uncertain and may depend on how responsibilities are defined in contracts between project actors.	Contractor 0; Industry Expert 1
Requirements for implementation	Need for knowledge and dialogue	Structural engineers need more knowledge and guidance to use ERC correctly.	Industry Expert 0; Contractor 1
		Concrete producers need clearer regulations and technical guidance for testing and documenting performance.	Concrete Supplier 1; Industry Expert 1
		ERC requires stronger dialogue between contractors and concrete producers.	Contractor 0; Contractor 3
	Increased need for testing and verification	The use of alternative binders and new binder combinations may increase the need for testing and verification.	Contractor 2
		A clear verification system is needed for assessing new binder combinations.	Contractor 3
	Industry capacity	Larger companies are better prepared to implement ERC while smaller ones might need further investment in laboratories and technical expertise.	Concrete Supplier 2; Concrete Supplier 0
Properties of the concrete affected	Workability	Workability is essential for site acceptance of climate-optimized concrete and smooth transition.	Concrete Supplier 0
		The effect on workability depends on binder type and replacement level.	Concrete Supplier 1
	Increased water/cement ratio	ERC may allow higher water/cement ratios if durability performance is verified, which may improve workability.	Concrete Supplier 3
	Strength development	Concrete mixes with alternative binders may have slower early-age strength development than conventional mixes.	Industry Expert 2
	Long-term strength	The long-term strength of mixes that contain alternative binders can still be satisfactory or even higher than ordinary Portland cement concrete.	Industry Expert 2; Concrete Supplier 0
Status of development and regulatory progress	Intention with ERC	To create a more harmonized and performance-based approach to concrete durability across Europe.	Industry Expert 0
		ERC shifts the focus from concrete composition to performance verification.	Industry Expert 0
	Regulatory barriers	ERC could not be harmonized across Europe due to the difficulty of reaching agreement between durability experts from different national contexts.	Industry Expert 1, Industry Expert 0
	Concrete report	The concrete report is being developed by industry experts to describe how ERC could be implemented in practice and to provide guidance for the Swedish concrete sector.	Industry Expert 0; Industry Expert 1; Industry Expert 2
	Future implementation pathway	The concrete report is seen as a bridge toward future ERC implementation in Sweden.	Industry Expert 0; Industry Expert 2
		ERC implementation is expected to be gradual, with a possible start as an optional performance-based route.	Industry Expert 0; Industry Expert 1

4.1 Benefits

The first main benefit identified in the interviews was that ERC may contribute to reducing the climate impact of concrete. Several interviewees connected ERC with the possibility of using concrete mixes with lower cement content and higher proportions of alternative binders, provided that the required durability performance can still be verified. As one contractor explained, *"the concrete manufacturers will be more free to use alternative binders in a higher rate. So, a transition to exposure resistance classes would mean that in many cases we can have a lower climate impact on the concrete that we use in our projects"* (Contractor 0). This indicates that ERC is not only seen as a technical classification system, but also as a tool that could support

the transition toward climate-optimized concrete. Similar views were expressed by a concrete supplier, who stated that the main environmental benefit is that exposure resistance classes could allow concrete with less cement and other mix compositions to be used (Concrete Supplier 1). In addition to the climate benefits, one contractor also mentioned that allowing a wider range of water-cement ratios for specific construction parts could be beneficial from both environmental and economic perspectives (Contractor 0). Therefore, the interviews suggest that ERC may help reduce unnecessary prescriptive limitations and give concrete producers more flexibility to design mixes that meet durability requirements with a lower climate impact.

The reduction in climate impact of the concrete could support the Swedish concrete construction sector in reaching its long-term climate goals, including the goal of climate-neutral concrete by 2045. By allowing concrete mixes to be specified based on the performance properties required for a specific project, ERC may reduce unnecessary margins and give concrete producers greater flexibility to develop mixes that satisfy durability requirements without relying only on prescriptive deemed-to-satisfy limits. This perspective was shared by both contractors and concrete suppliers. As one contractor stated, *"If we look at the roadmap to zero emission, I think this is a deal breaker to reach the goals that you can go from this traditional way with the exposure classes to more see what kind of properties the concrete needs to have"* (Contractor 2). This was also supported by a concrete supplier, who considered the environmental potential of ERC to be connected to broader green development in the concrete sector (Concrete Supplier 2).

Another way in which ERC may contribute to environmental goals is by supporting the use of alternative binders and reducing the cement or clinker content in concrete mixes. One contractor connected future climate targets directly to the increased use of alternative binders, stating that *"when it comes further on like, 2030 and beyond, I do think that in order to use concrete and achieve the goals that we have set up, all concrete has to be a quite high degree of alternative binders"* (Contractor 3). This indicates that ERC is perceived not only as a durability classification system, but also as a possible tool for enabling lower-carbon concrete solutions, provided that the required performance can be verified.

The benefit of enabling low-carbon concrete solutions was a perspective that representatives of all three interview groups shared together. One contractor stated that the ERC concept enables more freedom when choosing the concrete mix, as long as the requirements for durability are fulfilled (Contractor 0). As one industry expert stated, *"So today, I mean, we are using much, much more other things than Portland cement, and we have to manage this. Yes, and it's a necessity for the concrete business."* (Industry Expert 1). This means that the concrete industry is already providing projects that use climate-efficient concrete and that ERC is a necessity to ease the process, moving from prescriptive to performance-based methods of defining the concrete mix. This perspective was supported by a concrete supplier as well, where they argued that the current prescriptive methods include a lot of restrictions, and that ERC will be an opportunity to move forward by benefiting from the use of SCMs in the concrete mix (Concrete Supplier 2).

As shown in the frame of reference chapter, low-carbon concrete solutions can be achieved by replacing the cement or clinker content with the use of alternative binders. As the availability of some binders is decreasing, a need to find new binders will rise. One industry expert mentions that one of the the starting points of the ERC concept in Europe was to make it easier to introduce new binders (Industry Expert 0). ERC allows the concrete producers or binder providers

to test different concrete mixes that contain binders the industry does not have experience with, it may be easier to understand the properties of these new mixes. As one industry expert stated, *"It could be, uh, a way, an easier way to introduce a new type of materials that we don't have, uh, as much experience of."* (Industry Expert 2). One concrete supplier sees the benefit of introducing new binders as a business perspective, by allowing the suppliers to introduce binder combinations more quickly and offering customers better solutions (Concrete Supplier 3). One contractor stated, *"of course what we can do is optimize the structures more and be able to, for instance, use other materials than only cement or other new materials that will come in the future."* (Contractor 1). This perspective links the benefit of replacing the cement content in the concrete mix to the other benefit observed, the optimization of the structures.

Optimization is a concept that in this topic includes two aspects. One is to optimize the structures by removing or lowering the "unnecessary" margins, by reducing the cover thickness and therefore using less concrete in the structures. This aspect was not directly observed in the responses of the interviews, but is rather understood from the frame of reference. The other aspect is optimizing the concrete mix. As one industry expert stated, *"I, I think, uh, performance testing is good, then you can use the materials in the best way not to use uh more than you need, so that that that's I think an advantage.It can be used in that way, um."* (Industry Expert 2). One contractor sees this performance-based method as an opportunity to tailor the concrete mix to the specific needs of a structure (Contractor 2).

Businesses consistently evaluate the economic dimensions of any given innovation. Throughout the interview process and the analysis of the data it was observed that overall the concrete industry sees ERC as an opportunity to their profit, with little concerns. From the contractor's side, different perspectives appeared on how can the contractor profit by the use of ERC. As one contractor stated, *"if we were allowed to use a wider range of w/c ratio in a specific construction part, that would be beneficial for us, I believe, both in climate aspects and as well as economical aspects."* (Contractor 0). By optimizing the concrete mix and increasing the water-to-cement ratio, the price of the concrete would be decreased, because we are using less cement in the mix. Another economical benefit which is influenced by the optimization of the concrete mix is the reduction of cracks. Contractor 1 stated that by using less cement in the concrete mix, the structure is less affected by shrinkage and therefore less cracks appear, making the treatment process of cracks less costly.

Although contractors believe that methods to decrease the cement content in concrete will decrease the price of the concrete, which is theoretically true, suppliers see the optimization possibilities as an opportunity to increase the prices of these newer mixes of concrete. As one industry expert, who is also involved in the concrete supplying business stated, *"and then we can have a higher price also because we sell like a performance, not, not, not just a concrete, the gray mass, we sell something different, a more sophisticated product. That's interesting."* (Industry Expert 0). It is worth noting that the industry understands the possible increase of the need for tests and the price of the binders can influence the price of the concrete (Concrete Supplier 2). Therefore, more evaluation of this benefit will be required after the standards officially introduce ERC in Sweden.

As with all innovations, some projects will be the ones that benefit more or will be the first to benefit from the use of ERC. This section is not a benefit in its own, but rather intended to showcase these benefits of the topic in practical use. A consistency in perspectives between the interviewees was noted, where participants mentioned the structures exposed to chlorides will be

the ones to benefit most. In practice, examples of these structures are garages, swimming areas and buildings close to the sea (Contractor 3 & 0, Industry Expert 1 & 2, Concrete Supplier 2). The reason why these types of project will benefit the most, or will be the first to benefit, is that this class of exposure is the easiest step to introduce the method. This can be supported by *"Chloride is, is the one that we think and discuss the most because it's the easiest step."* (Contractor 3).

Regarding the size of the projects, this is where concrete suppliers stated that larger projects will benefit first. As one industry expert, who is also involved in concrete manufacturing stated, *"...in, in the beginning, when this concept is introduced into the standards, it will be, I think, mainly to really large projects."* (Industry Expert 0). This belief is motivated from the reason that introducing a new concrete mix for a specific project requires tests, knowledge, and therefore more expenses, which might not be covered if they were to be used in a small project.

4.2 Barriers, risks, uncertainties

The most supported barrier identified in the interviews is the multi-dimensional lack of knowledge regarding ERC. This finding also links directly to the Frame of Reference, where ERC is presented as a shift from prescriptive exposure-class requirements toward a more performance-based system. Such a shift requires knowledge not only of the ERC concept itself, but also of material properties, durability models, testing methods, fresh concrete behavior etc. However, the interviews indicate that this knowledge is not yet widely established in the industry. Apart from industry experts directly involved in ERC development, both contractors and concrete suppliers often had limited familiarity with the concept before the interview. As stated by a contractor, *"...I do know about the intention, uh, but I don't know the detail exactly how it works."* (Contractor 3), while one concrete supplier explained that *"I wasn't that into it before, uh, you mentioned it for me, and then I, but then I have read about it."* (Concrete Supplier 1). This suggests that although ERC is technically developed in the regulatory and scientific context, its practical understanding among the actors who will apply it is still limited.

A second dimension concerns structural engineers and prescribers, who are one of the most important actors in choosing and specifying concrete. Several interviewees described designers as slow to adopt new methods. One contractor stated that *"I think this is very new to everybody. And the designers are the last ones on the train."* (Contractor 1), while an industry expert similarly noted that structural engineers are generally slow to adapt to new things (Industry Expert 0). Concrete suppliers also described prescribers as often repeating familiar solutions instead of actively engaging with new performance-based alternatives (Concrete Supplier 2). This is important because the Frame of Reference shows that ERC requires a more active understanding of the relationship between exposure conditions, concrete resistance, and verification methods. If designers and prescribers lack this knowledge, the implementation of ERC may be delayed or applied in a conservative way that limits its intended benefits.

The lack of knowledge also has practical consequences during production. As stated by one contractor, *"If we have the people on site that are not engaged in this change, it's an increased risk that the casting will be faulty, that we will have more errors."* (Contractor 3). This connects the knowledge gap to the fresh concrete properties discussed in the Frame of Reference, since climate-optimized concretes and alternative binder systems may behave differently during transport, pumping, placing, compaction, finishing, and curing. Therefore, ERC implementation is

not only a design or regulatory issue, but also a production issue. Without sufficient education across the full construction chain, the transition to ERC may increase the risk of mistakes rather than support more reliable and optimized concrete solutions.

Finally, the interviews indicate that the knowledge gap may affect smaller companies more strongly than larger ones. Industry experts noted that major contractors and concrete producers tend to have more resources and material expertise, while smaller companies may lack the capacity to understand and apply new requirements related to ERC and alternative binders. This suggests that the implementation of ERC may create uneven conditions in the industry, where larger companies are better prepared to adapt while smaller actors may need additional guidance, education, and standardized procedures. This argument connects the lack of knowledge to another barrier observed throughout the interviews, the struggle of smaller companies to comply to regulations and implement ERC into their workflow.

This barrier is closely connected to the broader lack of knowledge discussed above. Smaller companies may be more affected by the transition to ERC because they often have fewer technical resources, less internal expertise, and more limited capacity to adapt to new regulatory requirements. As one industry expert stated, *"smaller producers, I think that they are, they will have problems with this concept because it's something completely new for them."* (Industry Expert 0). This issue was not only connected to concrete producers, but also to smaller contractors. As one contractor from a larger construction company explained, *"...we will be able to implement this, but not all small contractors that still have to fulfill the regulations."* (Contractor 0).

Besides the lack of knowledge, smaller companies may also struggle with the practical requirements connected to testing and verification. This is particularly relevant for smaller concrete producers, since ERC may require more testing to demonstrate concrete performance. If companies do not have their own laboratories, the process may become more difficult and costly. As one concrete supplier stated, *"...if we need to test. I, I think that, I mean, it would be easier for bigger companies to handle. It would be a bit more advanced probably."* (Concrete Supplier 3). Similar concerns were also expressed by Concrete Supplier 2 and Industry Expert 1. These statements suggest that the implementation of ERC should consider differences in company size and technical capacity. Otherwise, the system may be easier for large companies to adopt, while smaller actors may face greater difficulties in complying with the new requirements.

By combining the novelty of this topic, the introduction of new materials to the concrete mix, the possible need for laboratory testing, and the present lack of knowledge, it is expected that some level of complexity will be present, at least in the beginning of the implementation process. Representatives from all interview groups shared similar opinions regarding this barrier, although they approached it from different perspectives. Contractor 0 linked the diversity of products to this complexity, stating that *"we will have a greater range of products at our construction sites that are performing differently from each other"* (Contractor 0). This indicates that the use of a wider range of concrete products may become a challenge on construction sites, since different mixes may behave differently during production and casting. Concrete Supplier 2 linked the complexity to the lack of knowledge, stating that *"it's making it quite complex because and the knowledge about it has to be there as well."* (Concrete Supplier 2). Similarly, Industry Expert 0 stated that *"Already now it's complicated."* (Industry Expert 0). These statements suggest that the implementation of ERC may not only introduce new technical requirements, but also require a higher level of knowledge and coordination between the actors involved.

Another aspect of this complexity is related to the introduction of new technology and materials into the market. As Contractor 2 stated, *"It's always quite hard in the beginning to come with a new technology or a new material because you have to have a, a certain demand to get the prices down"* (Contractor 2). This suggests that ERC implementation may also create complexity from an economic perspective, especially in the early stages before demand increases and the products become more established. Therefore, while ERC may provide more flexibility in the long term, the transition period is expected to be more complex due to product diversity, knowledge requirements, testing needs, and market adaptation.

A further barrier identified in the interviews is the long transition time required before ERC can be used in practice. Interviewees emphasized that the implementation of ERC will not happen immediately, since the industry needs time to adapt its regulations, routines, testing capacity, and working methods. One contractor stated that *"in the moment we're not going to use [ERC] because it will be a very long transition time."* (Contractor 1), while an industry expert estimated that the transition may take *"10 years perhaps."* (Industry Expert 0). This indicates that, even if ERC is seen as technically promising, its practical implementation is limited by the time needed for the industry to build confidence, develop common procedures, and align the different actors involved. As one interviewee summarized, *"things are not going to happen overnight"* (Concrete Supplier 2). Therefore, the long transition period can be understood as an organizational and regulatory barrier, rather than only a technical one.

The final barrier identified in the interviews was the skepticism of clients toward new concrete technologies and binder systems. In this context, clients refer to the actors ordering or procuring the project, such as developers, private owners, public authorities, or other project owners. Since clients were not directly interviewed in this study, their perspective was mainly understood through the experiences of contractors and concrete suppliers. As one contractor stated, *"I heard this argument from some of the clients, so they were very hesitant to try the new concretes because, they were more comfortable with the old ones that they knew worked"* (Contractor 2). This suggests that the transition may take more time, since clients are key actors in the procurement process and need to be willing to accept new concrete solutions in their projects. This hesitation also reflects the conservative nature of the construction industry, where new materials and methods often need to be tested, verified, and trusted before they are used on a larger scale. This was also supported by a concrete supplier, who described the construction sector as a conservative business (Concrete Supplier 0), and by a contractor who stated that some clients do not believe in supplementary cementitious materials (Contractor 3). Therefore, client skepticism can be understood as a barrier that is closely connected to trust, experience, and confidence in the performance of new concrete solutions.

Client skepticism is also linked to other barriers identified in the interviews. One important connection is the lack of knowledge about available methods for defining concrete mixes and about the role of different binder systems. If clients do not understand how new concrete technologies are verified, or how alternative binders can still satisfy durability and performance requirements, they may be less willing to accept them in projects. This skepticism is also connected to the long transition time required for ERC implementation. The more hesitant clients are toward new methodologies, the longer it may take for these methods to become accepted in practice.

4.3 Changes in the industry

When new methods that influence several aspects of the concrete construction industry are introduced, changes in the industry can be expected. One of the themes identified in the interviews was therefore related to the changes that ERC may create in the market. The first change concerns increased competitiveness, especially between concrete suppliers. As mentioned by one contractor, *"There will be more competitiveness because you have a bigger freedom to change the recipes."* (Contractor 2). This indicates that concrete suppliers may have greater freedom to adjust their concrete mixes and deliver products that are more specifically adapted to the contractor's needs. As a result, competitiveness between concrete suppliers may increase, particularly in relation to the development of performance-based concrete mixes. This perspective was also supported by one concrete supplier, who stated that ERC could become a competitive advantage if a supplier is able to develop a concrete mix that others do not have (Concrete Supplier 1). This suggests that ERC may create opportunities for suppliers with strong technical knowledge, testing capacity, and experience with alternative binder systems to differentiate themselves in the market.

Another aspect of increased competitiveness concerns contractors. One contractor linked competitiveness to the ability of companies to deliver projects with lower costs and lower carbon impact (Contractor 0). In this sense, ERC may influence competition not only between concrete suppliers, but also between contractors who are able to use performance-based concrete solutions efficiently in their projects. Contractors that can offer lower-carbon solutions while still satisfying durability and performance requirements may gain an advantage in projects where environmental performance is an important criterion.

A further potential effect is that ERC may create opportunities for new suppliers to enter the market. One contractor suggested that in areas with high population density or extensive construction activity, new suppliers may emerge if there is demand for more specialized or performance-based concrete solutions (Contractor 2). This indicates that ERC could influence not only competition between existing suppliers, but also the structure of the supplier market itself. However, this development may depend on the technical capacity of each actor. Suppliers that are able to test, verify, and document performance-based concrete mixes may gain a stronger competitive position, while actors with limited resources may find it more difficult to compete.

The next change in the industry identified in the interviews was the potential shift in responsibilities between the main actors in the construction process. The interviews suggest that structural engineers will still have an important role in prescribing the concrete cover and the exposure resistance class. As one industry expert stated, *"...it will make the design firms, it will make their job much easier because they only have to decide what cover do I need and what kind of, uh exposure resistance class"* (Industry Expert 1). This indicates that the designer's role may become more focused on defining the required performance level, rather than prescribing the detailed concrete composition.

However, some interviewees indicated that more responsibility may move to the concrete producers. As one contractor stated, *"The responsibility for fulfilling the exposure resistance classes means more responsibility on the manufacturers of the concrete."* (Contractor 0). Since ERC focuses on performance rather than fixed recipes, concrete producers may become more responsible for developing, testing, and documenting mixes that fulfill the required exposure resistance class. This is also supported by an industry expert, where they mention that is the responsibility

of the concrete producers to fulfill the requirements for durability (Industry Expert 1).

One interesting perspective was shared by a concrete supplier, who stated that *"...so perhaps we're looking at a new type of prescribers, prescribers who also have knowledge about this."* (Concrete Supplier 2). This suggests that ERC may create a need for actors with stronger knowledge of concrete materials and performance-based requirements, who can support the dialogue between structural engineers and concrete producers.

As the responsibilities of each actor may shift, uncertainties may also arise regarding who is liable if performance fails. This was one of the codes identified in the interviews. As one contractor stated, *"...it would be harder to say who is the actual responsible, who has the actual responsibility."* (Contractor 0). Since ERC is still at an early stage of implementation, several aspects related to responsibility and liability remain uncertain. Therefore, clearer regulations, contractual arrangements, and role definitions will be needed to clarify how responsibility is distributed between designers, contractors, and concrete producers.

However, other interviewees emphasized that liability will also depend on the contracts signed between the parties (Contractor 2; Industry Expert 1). One industry expert further stated that if the concrete fulfills the requirements set by the prescriber and the relevant regulations, it may be difficult to place liability on the concrete producer (Industry Expert 2). This suggests that the introduction of ERC may require not only technical verification methods, but also clearer contractual and legal frameworks for assigning responsibility when failures occur.

4.4 Requirements for implementation

As discussed in Section 4.2, lack of knowledge was one of the main barriers identified by the interviews regarding the implementation of ERC. Therefore, it was expected that the need for increased knowledge would also appear as an important requirement for implementation. One recurring argument, supported by contractors, industry experts, and concrete suppliers, was that structural engineers need more knowledge about the concrete they specify. This is important because ERC requires the prescriber to define performance requirements rather than relying only on traditional prescriptive rules. As one industry expert stated, *"we have to have a lot of education and to really teach, especially those who prescribe. How to prescribe according to this system."* (Industry Expert 0). Another industry expert similarly argued that some form of guidance will be needed for structural engineers (Industry Expert 1). This indicates that education, guidance documents, and practical support may be necessary in order for prescribers to understand how to use ERC correctly.

A similar perspective was shared by concrete suppliers, who emphasized that clearer regulations and guidance are also needed on the producer side. One concrete supplier described regulations as the equipment needed in their process, since they would provide a clearer basis for testing different recipes and understanding what needs to be done. This suggests that knowledge is not only needed by designers, but also by concrete producers, who will have to develop, test, and document mixes that fulfil the required exposure resistance class (Concrete Supplier 1).

Another requirement identified within this code was the need for increased dialogue between contractors and concrete producers. Since ERC gives more freedom in how concrete recipes are developed, communication between the actors becomes more important to ensure that the right product is delivered to the construction site. As one contractor stated, *"I think there will*

be a need for an increased dialogue between us contractors and the concrete manufacturers in order to receive the right products on our construction sites." (Contractor 0). This was also supported by another contractor, who emphasized the importance of a good relationship with the supplier and the need to receive enough information before changes are made, especially for more critical structures (Contractor 3). However, the quality of communication between actors may also depend on the technical capacity within the companies involved. As one industry expert stated, *"I think it depends on the contractor actually. If it's a big contractor that has a technical staff behind, then it's easier to have a good communication."* (Industry Expert 0). This links the need for dialogue to knowledge capacity, since larger companies with dedicated technical staff may be better positioned to communicate with concrete producers and understand the implications of changes in concrete mixes. Smaller companies, with less internal technical support, may face greater difficulties in this process. This connection between knowledge and dialogue also supports why these two requirements were treated within the same code during the analysis.

The interviews therefore suggest that implementation of ERC will require both increased technical knowledge and stronger communication between the actors involved. Structural engineers need to understand how to prescribe concrete according to the ERC system, while concrete producers need sufficient guidance to test and verify concrete performance. At the same time, contractors and producers need a closer dialogue to translate project requirements into suitable concrete products. This makes knowledge and dialogue central requirements for ERC implementation.

Another requirement for the successful implementation of ERC in Sweden, which is also linked to the theme *Changes in the industry*, was the increased need for testing. With the use of alternative binders and the possible introduction of new binder combinations in the future, testing is needed to understand how new concrete mixes behave and whether they fulfill the required performance criteria. This was supported by several interviewees. For example, one contractor stated, *"you have to test different or new mixes to see if you can work with the concrete."* (Contractor 2).

The increased need for testing may also create higher initial costs for concrete producers. One concrete supplier stated, *"So it, it, it will require time and advanced lab equipment, for example, uh, so it would be higher initial costs."* (Concrete Supplier 1). These higher initial costs can help explain why some interviewees believed that larger projects will be the first to benefit from ERC. In larger projects, the initial cost of testing and mix development may be easier to justify, since it can potentially be balanced by savings from optimized concrete mixes, lower cement content, and reduced climate impact.

Another important point was the need for a clear system to verify new binder combinations. One interviewee explained that when new SCMs or cement types are introduced, there must be a system to ensure that the new combination has the required properties (Contractor 3). This suggests that testing alone is not sufficient, the results also need to be connected to accepted procedures for verification and approval.

However, the interviewees did not all share the same view regarding the extent of testing required. One concrete supplier argued that testing may not become excessive if certain binder combinations can be qualified in advance, similar to existing approaches: *"if you can show that if it's enough that the cement company, for instance, qualify a certain binder combination like*

they do today with they can do it with the EPCC concept. If you do it the same way, then it will not be a lot of testing." (Concrete Supplier 3). This suggests that the testing burden is dependent on how the ERC system is implemented, and whether standardized or pre-qualified binder combinations are accepted.

The final requirement for the successful implementation of ERC was the need to assess the existing capacity of the Swedish concrete industry and identify what needs to be improved. The interviews indicate that performance-based approaches are already used to some extent in Swedish concrete projects. As one contractor stated, *"even now it's possible to use function testing in order to do something similar to ERC. There are possibilities within the standards, but it's not obvious how it's done and how to do it properly in projects, which means that we do put a lot of hope in the ERC that when it comes it will be easier, but the question is also then how is it implemented."* (Contractor 3). This suggests that the industry already has some experience with performance-based verification, but that clearer guidance is needed for how such methods should be applied in practice. In this sense, ERC is expected to provide a more structured framework for moving from prescriptive requirements toward performance-based durability design. Several studies have investigated the use of alternative binders to ordinary Portland cement and their influence on concrete properties, some of which are presented in Chapter 2. The interview findings indicate that this knowledge is not limited to scientific or laboratory contexts, but is also beginning to support practical applications in real construction projects.

Another important aspect of industry capacity is the readiness of different companies to implement ERC. This issue is also linked to the barriers discussed earlier, where smaller companies were expected to face greater difficulties. Several interviewees argued that larger companies are better positioned to implement ERC because they often have more technical staff, experience, and testing capacity. As one concrete supplier stated, *"if you want to develop the concrete you have to, it's good to have the capability yourself."* (Concrete Supplier 2). This indicates that concrete producers with their own testing capacity may be able to develop and verify new concrete mixes more efficiently. Having internal laboratory capacity can make the process faster and easier, while companies without such capacity may become more dependent on external testing and support.

At the same time, the interviews suggest that the existing capacity may not be sufficient in all areas. One concrete supplier mentioned that they only have small laboratories and are not yet able to perform more production-based testing (Concrete Supplier 0). This indicates that successful ERC implementation may require not only knowledge and guidance, but also investment in testing facilities, production-scale verification, and technical competence across the industry.

4.5 Properties of the concrete affected

Throughout the interviews, concrete properties were frequently discussed by the interviewees, especially in relation to the replacement of ordinary Portland cement with alternative binders. The interviews indicate that this replacement affects the properties of concrete, especially the fresh properties that are important during production and casting. Among these, workability was one of the most frequently mentioned.

Workability was mainly discussed as a practical requirement for the successful use of climate-optimized concrete on construction sites. As one concrete supplier stated, *"So it have to be a*

similar workability, I think. To, to get a smooth transition towards climate optimized concrete, you, you, you can't have too, too much difference." (Concrete Supplier 0). This suggests that for new concrete mixes to be accepted and used efficiently on site, they should not behave too differently from conventional mixes. If the workability changes significantly, the established working methods for transport, pumping, placing, compacting, and finishing may also need to change. This increases the risk of execution problems, especially when site workers are unfamiliar with the new material behavior.

The interview findings also show that changes in workability depend both on the type of SCM or alternative binder used and on the replacement level. One concrete supplier explained that small replacement levels may not be noticed on site, while higher replacement levels can change the behavior of the concrete: *"Yeah, both good and bad, I could say, yeah, in the beginning when we started and we took just 10% with a different binder. Uh, they, they didn't, uh, they didn't actually felt something. It was just like the ordinary, but as we go further and further, like 20% and 30% and stuff like that, things happen, and they notice that the, the product is not the same."* (Concrete Supplier 1). This indicates that the transition towards lower-clinker concrete may be relatively simple at low replacement levels, but more demanding when the binder composition is changed more substantially. This was also supported by one contractor, who mentioned that complaints from construction sites had occurred regarding workability and stickiness in some binder systems (Contractor 3).

Another factor discussed by the interviewees was the water/cement ratio. This code is closely connected to workability, but it also shows how the use of alternative binders may change the way concrete mixes are designed. Several interviewees suggested that, if the same durability resistance can be achieved with a lower cement content and a higher water/cement ratio, the concrete could become more workable. As one concrete supplier stated, *"if you think about the ERC concept we, I would like to see that we go towards higher water/cement ratios, which also means that we get a more workable mix usually."* (Concrete Supplier 3).

This was also supported by one contractor, who explained that concrete producers have shown examples where the cement content could be reduced from 380 kg/m^3 to 300 kg/m^3, while still achieving the same resistance class as the previous concrete (Contractor 1). The same contractor also mentioned that, for parking decks, Heidelberg and Thomas Betong have shown that a concrete with a water/cement ratio of 0.5 together with GGBS can achieve similar chloride resistance as a concrete made only with cement and a water/cement ratio of 0.4. This indicates that the use of alternative binders, together with a performance-based classification such as ERC, may allow concrete mixes to be optimized beyond the traditional prescriptive limits. This could reduce cement content and climate impact while still maintaining durability performance. At the same time, the higher water/cement ratio could improve workability, which would make the concrete easier to handle on site.

Another important property discussed by the interviewees was early-age strength development. The most common view was that concrete mixes containing SCMs or alternative binders generally develop strength more slowly than conventional Portland cement mixes. As one industry expert stated, *"Yeah, all this supplementary cementitious materials slow down the, the strength development, yeah. Sure. Uh, most of them, at least that we use today, so."* (Industry Expert 2). This argument was supported by other concrete suppliers too. If the concrete reaches the required strength later, formwork may need to remain in place for longer, which can slow down the project and increase costs.

Temperature was also mentioned as an important factor. One concrete supplier explained that mixes containing slag may perform sufficiently during summer, but that short-term strength development becomes slower during winter (Concrete Supplier 0). This indicates that the use of alternative binders is not only dependent on the binder type and replacement level, but also on the casting conditions. Therefore, the interviews suggest that early-age strength development is one of the main practical challenges when alternative binders are used, particularly in projects where construction speed, formwork cycles, and cold-weather casting are important.

Although the interviewees mainly described SCMs and alternative binders as challenging for early-age strength development, they also stated that the long-term strength can be equal to or higher than that of conventional Portland cement concrete. As one industry expert explained, *"If you, if you think about these SCMs, often they have uh, higher long-term strength. Uh and then uh and then, compared to Portland cement."* (Industry Expert 2). This was supported by one concrete supplier, who stated that strength is usually not the main issue, and that mixes with slag may continue to gain strength after 28 days: *"Uh, my view is that you get uh, a, higher hard strength. When when you go with slag when you go past 28 days, you still have an increase."* (Concrete Supplier 0). This indicates that while alternative binders may slow down early strength development, they can still provide satisfactory or improved long-term strength. Therefore, the main challenge is not necessarily the final strength of the concrete, but the slower development of strength during the early construction phase.

Overall, the interview findings are consistent with the Frame of Reference. The literature states that fresh concrete must be designed not only for hardened properties such as strength and durability, but also for fresh-state requirements such as workability, stability, pumpability, setting behaviour, and early-age strength development [AB Svensk Byggtjänst, 2017]. This supports the interviewees' concern that climate-optimized concrete must remain practical to transport, place, compact, and finish on site. The Frame of Reference also shows that the effect of alternative binders is material-specific: GGBS can improve flowability and pumpability, while silica fume may reduce workability as the replacement level increases [Hanzlah Akhlaq et al., 2025, M. Mazloom et al., 2004]. This corresponds with the interviewees' view that the behaviour of the concrete depends on both binder type and replacement level. Regarding strength, the literature also supports the interviewees' distinction between slower early-age strength development and potentially satisfactory or improved long-term strength, especially for mixes containing slag. Therefore, both the interviews and the literature indicate that alternative binders can enable lower-clinker concrete, but their use requires careful control of workability, water/cement ratio, early strength development, and casting conditions.

4.6 Status of development and regulatory progress

Until now, the results discussed in this chapter have focused mainly on the interviewees' perspectives on ERC and how these perspectives relate to the information presented in Chapter 2. The potential benefits and barriers have been discussed, followed by the expected changes in the industry, the requirements for implementation, and the concrete properties affected by the use of alternative binders. However, most of these themes are based either on findings from the Frame of Reference or on or opinions of the experts based on their knowledge. Therefore, it was also necessary to examine the current status of ERC development in Sweden through interviews with industry experts who are directly familiar with the process. This final theme addresses the development and regulatory progress of ERC. It covers the background and original intention

of ERC, the reasons why full harmonization across Europe has been difficult, the current state of development in Sweden, and the expected next steps for implementation.

The interviews indicate that one of the main intentions behind ERC was to create a more harmonized and performance-based approach to concrete durability across Europe. Several industry experts explained that the current durability requirements differ between countries, and that ERC was intended to reduce the need for separate national provisions. As one industry expert stated, *"the meaning was that with this system, all these national provisions, they should be removed"* (Industry Expert 0). This was also supported by another expert, who confirmed that ERC was partly intended as a form of harmonization across Europe, since durability rules are currently different between countries (Industry Expert 1). In this sense, ERC can be understood as an attempt to create more consistent durability requirements based on concrete performance rather than mainly on national prescriptive rules.

Another intention identified in the interviews was to move the focus from fixed concrete composition toward testing and verification of performance. One industry expert explained that the intention was that EN 206 should not describe how concrete should be composed, but rather should address how concrete with different new mixes can be tested using different test methods (Industry Expert 0). This suggests that ERC is not intended to prescribe the concrete recipe itself, but to provide a framework for verifying whether the concrete fulfills the required durability performance. Similarly, another expert described ERC as a structural durability problem rather than a direct concrete composition issue, meaning that the system primarily concerns durability requirements and verification rather than execution on site or fresh concrete handling (Industry Expert 0).

The interviews also show that ERC was intended to support innovation while still protecting durability. One industry expert emphasized that sustainability goals should not override durability requirements, since environmental and durability aspects are closely connected (Industry Expert 1). This indicates that ERC is not intended to simply allow lower-carbon concretes without control, but rather to make it possible to use new binder systems and concrete mixes when their performance can be demonstrated. In this context, the producer of the binder or concrete may need to take greater responsibility for testing and documenting performance, as one expert stated that the producer of the binder should test their binder (Industry Expert 0).

Finally, ERC was described as a concept that can be used in different ways. It can be applied directly by testing concrete performance instead of relying only on requirements such as maximum water-to-binder ratio or minimum cement content. However, one industry expert also noted that the ERC concept could be used to develop new deemed-to-satisfy rules (Industry Expert 1). This suggests that ERC is not necessarily a complete rejection of prescriptive approaches, but rather a framework that can support both direct performance verification and improved prescriptive rules based on tested durability performance. By also having new deemed-to-satisfy rules, the smaller companies that cannot have the capacity to do the testing themselves, can still function without struggling to comply.

The interviews indicate that one important reason why ERC could not be fully harmonized across Europe is the difficulty of reaching agreement between durability experts from different national contexts. As one industry expert stated, *"when you come to especially durability experts, there are sometimes too many experts and everybody has his own model and his own thinking [...] some of them are very strong-minded"* (Industry Expert 1). This suggests that the development

of ERC has been influenced by differing technical traditions and expert opinions, where national specialists may prefer models and methods that are already established in their own context.

As discussed in Section 2.6, calculation models are expected to support the development of tabulated values. However, the interviews suggest that agreement on which models should be used has been difficult to reach, particularly because different experts and countries may consider their own models to be the most reliable. According to another industry expert, the largest differences between the models are found for structures exposed to chlorides (Industry Expert 0). The same interviewee explained that comparisons between different models showed that some approaches may underestimate chloride ingress, which may lead to an overestimation of durability and service life: *"they see that this model [...] will underestimate the chloride ingress. And that is not really good because then we overestimate the durability, we overestimate the service life"* (Industry Expert 0). This helps explain why the implementation of ERC is currently progressing through national approaches rather than through one fully harmonized European model. The findings therefore show that regulatory progress depends not only on the formal introduction of ERC in the new Eurocode, but also on the development of accepted calculation models, verification methods, and national rules that can translate the concept into practical design guidance.

The need for national rules to implement ERC in Sweden brings us to the next code, which concerns the concrete report. This report, as described by the industry experts, is a proposal on how experts think ERC should be implemented. As one industry expert described it, *"it will be like a description of the concept from the beginning. In the first chapter, and then we will talk a little about these different models to make predictions. Then we will talk about how to classify different concrete based on testing. We will have a part that is about how to prescribe concrete according to this exposures resistance class system. There will be a little about how the contractor should work. And then of course it could be a future research."* (Industry Expert 0). This means that the concrete report will cover the whole topics regarding ERC from the early stages of the project when concrete is prescribed until the concrete reaches the construction sites. Another industry expert added that the report may also give guidance on the types of the materials that can be used (Industry Expert 2). Experts stated that the report should be published by the end of 2026 (Industry Expert 0).

Although this report covers most of the crucial aspects of the topic, experts stated the published version will be a first one and that it should be revised when they have better knowledge (Industry Expert 0). This means that the lack of knowledge is the biggest barrier the topic is currently facing and that it is neccessary to close this gap, in order for ERC to be implemented in a real project as soon as possible.

The need for national guidance for the implementation of ERC in Sweden leads to the next code, which concerns the development of a concrete report. According to the interviewed industry experts, this report is intended to describe how ERC could be implemented in practice and to provide guidance for the Swedish concrete sector. As one industry expert explained, *"it will be like a description of the concept from the beginning. In the first chapter, and then we will talk a little about these different models to make predictions. Then we will talk about how to classify different concrete based on testing. We will have a part that is about how to prescribe concrete according to this exposures resistance class system. There will be a little about how the contractor should work"* (Industry Expert 0). This indicates that the report is expected to cover several stages of the ERC implementation process, from the basic description of the concept to

durability prediction models, testing, classification of concrete, prescription of concrete according to exposure resistance classes, and practical guidance for contractors. Therefore, the report can be seen as an important step in translating ERC from a regulatory concept into practical guidance for design, production, and construction. Another industry expert also stated that the report may provide guidance on which types of materials can be used in relation to ERC (Industry Expert 2). According to the interviews, the report is expected to be published by the end of 2026 (Industry Expert 0; Industry Expert 1).

However, the experts also emphasized that the first version of the report should not be seen as a final solution. One industry expert stated that *"this is like a first version of the report and that it should be revised when we have better knowledge"* (Industry Expert 0). This shows that the development of ERC in Sweden is still in a learning phase, where guidance is being prepared while practical experience, testing data, and industry knowledge are still developing. The need for better knowledge also connects this theme to several of the previous findings. The potential benefits of ERC depend on having reliable methods for verifying durability performance, while many of the barriers identified in the interviews are related to uncertainty regarding models, testing methods, responsibilities, and practical application. Similarly, the changes required in the industry and the properties of concrete affected by alternative binders cannot be fully addressed without further knowledge on how ERC performs in real projects.

Looking towards the future, the interviewed experts described the concrete report as an important step between the current development phase and the possible future implementation of ERC in Swedish standards and practice. Several experts indicated that its results could later form the basis for further standardization. As one industry expert stated, *"the results from this concrete report should be in some way implemented in the SS 137003"* (Industry Expert 0). This suggests that the report is not only intended as an explanatory document, but also as a possible foundation for future updates of Swedish concrete regulations. The experts also described the future implementation of ERC as a gradual process rather than an immediate regulatory change. One industry expert explained that the first step is to develop guidelines for how ERC can be applied, including tabulated values for different resistance classes and exposure conditions. The same expert stated that this would later go into the natural application of the Eurocode and, at some point, be introduced into the national application documents for concrete in EN 206 (Industry Expert 2).

However, the interviews also suggest that the future use of ERC will probably begin as an optional approach rather than as a mandatory replacement of the existing system. One industry expert stated that ERC will not be mandatory and that the old system with prescribed rules will still remain available (Industry Expert 0). Another expert similarly explained that, if several countries develop their own approaches during the coming years, this could provide a basis for future harmonization by allowing countries to compare experiences and learn from each other (Industry Expert 1).

Therefore, the experts present the future of ERC in Sweden as a stepwise development. In the present stage, the focus is on preparing the concrete report, developing guidance, and increasing knowledge. In the future, this guidance may be incorporated into Swedish standards and national application documents, while ERC may initially function as an alternative performance-based route alongside the existing prescriptive system. This shows that the implementation of ERC is expected to depend not only on regulatory decisions, but also on accumulated experience, practical testing, and confidence from the industry.

Chapter 5

Conclusions

This chapter presents the main conclusions of the study by firstly recommending actions that the industry can take, followed by answering each research question individually. It also identifies areas for future research based on the findings, limitations, and remaining uncertainties related to the implementation of ERC in Sweden.

5.1 Actions recommended for successful implementation of ERC

- Increase knowledge about ERC in contractors, structural engineers, concrete producers, and site personnel, since the current lack of knowledge is one of the main barriers to implementation.
- Develop clear guidance for structural engineers on how to prescribe concrete according to exposure resistance classes.
- Strengthen the dialogue between contractors and concrete producers to ensure that project requirements are met and that concrete mixes can be delivered and handled correctly on site.
- Establish standardized testing and modelling procedures for concrete mixes containing alternative binders, so that durability performance can be demonstrated in a reliable way.
- Assess and improve laboratory and technical capacity, especially among smaller concrete producers and contractors, since limited resources may make ERC more difficult to implement.
- Adapt construction-site routines to account for changes in fresh concrete properties, particularly workability, pumpability, placing, finishing, curing, and early-age strength development.
- Provide training for site personnel before using climate-optimized concrete mixes, as unfamiliar concrete behavior may increase the risk of casting errors and production problems.
- Begin implementation through pilot projects, preferably in larger projects or chloride-exposed structures where the benefits of ERC are expected to be most promising.
- Use early projects to build client confidence, demonstrate performance, and reduce skepticism toward new concrete technologies and alternative binder systems.

- Follow the development of the Swedish concrete report and future national guidance, since these are expected to provide the practical guidance for how ERC can be implemented in Sweden.
- Treat ERC implementation as a gradual transition rather than an immediate change, allowing the industry to build knowledge, testing experience, routines, and confidence over time.

RQ1 - What are the potential advantages that ERC brings in concrete construction projects in Sweden?

The benefits of shifting from a prescriptive to a performance-based approach are mainly environmental. By allowing prescribers greater freedom to use alternative binders to ordinary Portland cement, ERC enables concrete mixes with a lower carbon footprint. This method of prescribing the concrete mix can help the Swedish concrete industry come closer to its long-term climate goals. Economic benefits were also identified, where both contractors and concrete producers see ERC as an opportunity to reduce material costs and develop more competitive products.

RQ2 - What are the barriers, including uncertainties and risks, that currently limit the practical implementation of ERC in concrete construction projects in Sweden?

Several barriers introduce uncertainty for the main actors involved. The primary barrier is a multi-dimensional lack of knowledge. Most industry actors are unaware of the approaching change, structural engineers lack experience prescribing concrete based on performance, and concrete producers lack regulatory guidance for testing alternative binder mixes. Smaller companies face additional capacity constraints due to limited technical resources and laboratory equipment. Further barriers include increased complexity, a long transition period, and client skepticism.

RQ3 - How do the properties of the fresh concrete change and how do they affect the workflow?

The properties of the fresh concrete are affected when ordinary Portland cement is partially replaced with alternative binders. These changes have direct effects for the workflow on construction sites. The interviews indicate that workability is the most critical fresh property, as site workers need to be able to transport, pump, place, compact, and finish the concrete. At low replacement levels, the difference in behavior may not be noticeable, but at higher replacement levels the concrete can behave differently from what site workers are used to work with, which increases the risk of execution errors.

Early-age strength development is another property affected by the use of alternative binders, as most SCMs slow down strength gain compared to ordinary Portland cement mixes. This has practical consequences for construction speed, particularly in cold weather conditions where the effect is more noticeable. However, the long-term strength of mixes containing alternative binders can be equal to or higher than that of conventional concrete. This means that the main challenge is managing the early construction phase rather than the final structural performance.

These findings suggest that a successful transition to climate-optimized concrete requires not only technical adjustments to the mix design, but also adaptation of site practices, worker knowledge, and casting procedures.

RQ4 - What technical and organizational actions are required within a contractor's working process to implement ERC, to efficiently translate from a regulatory concept to practical use?

To successfully implement ERC in the Swedish concrete construction industry, actions on multiple directions are required. Education and guidance targeted at structural engineers and concrete producers would directly reduce the knowledge gap. Clearer regulations and standardized testing procedures would support smaller companies that cannot independently develop the required capacity. At the site level, the changes in fresh concrete properties discussed in Chapter 5.1 further highlight the need for trained site personnel who can handle the different behavior of climate-optimized concrete mixes. The increased use of alternative binders may also require investment in additional laboratory capacity, so that testing and verification of new concrete mixes can be carried out efficiently. Furthermore, as ERC shifts greater responsibility for mix development and performance verification toward concrete producers, the question of liability if performance fails will need to be clearly defined through the contracts signed between the parties involved. Client skepticism can be addressed gradually through demonstrated performance in early projects, particularly in chloride-exposed structures where the benefits of ERC were seen to be most promising. These findings suggest that while ERC presents an opportunity for both environmental and economic progress, the smoothness and speed of the transition will depend largely on how effectively the knowledge, capacity, and contractual gaps are addressed across the industry.

RQ5 - What is the status of regulatory progress in Sweden and when will ERC be implemented?

ERC was originally intended to create a harmonized, performance-based approach regarding concrete durability across Europe, shifting the focus from fixed concrete composition to testing and verification of performance. However, full harmonization could not be achieved due to disagreements between durability experts, especially on which calculation models should be used for chloride-exposed structures. Therefore, implementation is currently being developed on a national level.

In Sweden, the most significant development is the concrete report, currently being prepared by industry experts. This report is intended to describe how ERC can be implemented in practice, covering durability prediction models, testing and classification of concrete, prescription according to exposure resistance classes, and practical guidance for contractors. It is expected to be published by the end of 2026 and is seen as the bridge between the current development phase and future implementation in Swedish standards.

ERC implementation in Sweden is expected to be gradual. It will likely begin as an optional performance-based route alongside the existing prescriptive system, before eventually being incorporated into Swedish national application documents. These findings suggest that while the regulatory direction is clear, full practical implementation remains some years away and will depend on the knowledge and experience accumulated through

the concrete report and early adopter projects.

5.2 Future work

Based on the findings and limitations of this study, some areas for future research can be identified. Firstly, as mentioned in Chapter 3, the view from a client's perspective is not taken directly, but rather from interviews with actors that are in contact with clients. Future research could include how clients perceive ERC and what factors would increase their willingness to accept performance-based concrete solutions. The capacity of smaller concrete producers and contractors and what specific investments and support are needed for them can be an area where future research is needed.

Other future work that might be beneficial for the development of the topic can come after the publication of the concrete report, to investigate how can the concrete report be implemented into Swedish construction regulations. Finally, since contradicting views were identified regarding the economic implications of ERC, a quantitative study on how ERC affects concrete pricing and project costs would complement the qualitative findings of this study and provide a clearer picture of the economic impact of the transition.

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Appendix A

Interview questions

A.1 Contractor 0

1. Under what conditions would ERC increase flexibility for using low-climate concrete, and when would it not?
2. As a climate and construction specialist, in what ways does ERC help from a climate perspective and how do you view their future impact on reinforced concrete construction? Do you expect structures using the new mixtures of concrete to be more durable and at the same time more climate-efficient/friendly?
3. Which phases of the design and construction process do you expect to be most influenced by ERC?
4. Which phases of the design and construction process do you expect to be most influenced by ERC? What technical and organizational actions within the contractor organization are required to make ERC implementation as smooth as possible and minimize problems during this transition? What should be the first practical implementation step?
5. Which phases of the design and construction process do you expect to be most influenced by ERC? Do you expect a shift in responsibilities for selecting/specifying the right concrete mix for a structure? If yes, how do you think these shifts will occur?
6. Which phases of the design and construction process do you expect to be most influenced by ERC? Who do you expect to be the main actors making these decisions, and how should responsibility/liability be handled if something goes wrong (e.g., delayed strength, buildability failure)?
7. Which phases of the design and construction process do you expect to be most influenced by ERC? How could ERC positively affect the contractor's business?
8. Which phases of the design and construction process do you expect to be most influenced by ERC? Since ERC is performance-based, are there currently any projects or practices that already move away from prescriptive requirements and use a more

performance-based approach similar to ERC?

9. Which phases of the design and construction process do you expect to be most influenced by ERC? What potential risks should be considered regarding ERC and its implementation?
10. Which phases of the design and construction process do you expect to be most influenced by ERC? Which capabilities are already sufficient, and which are missing for successful ERC implementation in Sweden?
11. Which phases of the design and construction process do you expect to be most influenced by ERC? In your view, does Sweden currently have the capacity to implement ERC in practice?
12. Which phases of the design and construction process do you expect to be most influenced by ERC? Can you give an example of a project the contractor has worked on (or is working on) that would benefit from using ERC? Are there cases where ERC could do the opposite, meaning they would create more risks than values?
13. Which phases of the design and construction process do you expect to be most influenced by ERC? What is your overall view on the shift toward a more performance-based approach for designing and delivering future concrete structures?

A.2 Contractor 1

1. Which phases of the design and construction stage of the project will be most influenced by ERC? At which decision points does ERC change what the contractor must deliver or verify regarding documents?
2. What technical and organizational actions within the company are required to be made in order for the implementation of ERC to be as smooth as possible and not to have as many problems throughout this shift? What should be the first "practical implementation step"?
3. Will there be a shift in the responsibilities of deciding the right concrete mix for a structure? If yes, how would you predict these shifts are going to happen.
4. Who will be the main actors deciding and who will be liable if something goes wrong (delayed strength/ buildability failure etc.?
5. How could ERC positively affect the business of the company? Are there current projects using performance verification routes or project-specific durability testing that resemble ERC logic?
6. What are the potential risks that should be taken into account regarding this topic?
7. Which capacities are already sufficient, and which are missing regarding the successful implementation of ERC in Sweden? (Does Sweden currently have the capacity to implement ERC?)

8. Can you describe a project type where ERC clearly adds value, and one where ERC likely adds risk. What makes the difference?
9. What project types should be 'early adopters' of ERC, and what project types should be excluded at the start?
10. What is your overall thought on this shift towards a performance-based method of designing future buildings?
11. What are the main differences between laboratory behavior and real site conditions?
12. How would the involvement of the contractor in the working group of the ERC committee improve the quality work of the committee?
13. What is the Concrete Report and what is its purpose? What is included in the Concrete report?
14. When is the Report Being published? What are the main dependencies or unresolved topics that could delay it?
15. What are the barriers the committee is facing right now? How are they planned to be solved?

A.3 Contractor 2

1. What is your background, what projects have you worked with and what do you currently work with?
2. Can you walk me through the usual phases from early concept to issued-for-construction drawings, and what decisions are expected at each phase?
3. In your own words, what are programhandlingar, systemhandlingar, and bygghandlingar, and what concrete-related information normally appears in each?
4. How familiar are you with the ERC (performance-based method for concrete mix)?
5. Which phases of the design and construction stage of the project will be most influenced by ERC? At which decision points does ERC change what the contractor must deliver or verify regarding documents?
6. What technical and organizational actions within the company are required to be made for the implementation of ERC to be as smooth as possible and not to have as many problems throughout this shift? What should be the first "practical implementation step"?
7. Will there be a shift in the responsibilities of deciding the right concrete mix for a structure? If yes, how would you predict these shifts are going to happen.
8. Who will be the main actors deciding and who will be liable if something goes wrong (delayed strength/ buildability failure etc.?
9. How could ERC positively affect the business of the company?

10. What are the potential risks that should be taken into account regarding this topic?
11. Which capacities are already sufficient, and which are missing regarding the successful implementation of ERC in Sweden? (Does Sweden currently have the capacity to implement ERC?)
12. Can you describe a project type where ERC clearly adds value, and which kind of projects should be 'early adapters' of ERC?

A.4 Contractor 3

1. What is your background and what do you currently work with? How familiar are you with the ERC?
2. What are some activities/steps that the industry can follow to reduce the climate impact of the cement/concrete (to describe the cement activities)?
3. To what extent are those measures realistic in practice? Can all of them be implemented or should there be further changes in the industry workflow?
4. Do all these steps increase climate efficiency and reduce costs, or do some of these steps increase costs of the projects?
5. In what ways could the ERC help the industry to reach their goals of being climate neutral by 2045?
6. Which exposure classes would benefit most from ERC and in which ones would require more careful concrete mix design consideration? If not familiar with ERC and exposure classes: Which types of structures, environments, or exposure situations would benefit more from a performance-based mix of concrete?
7. Which fresh concrete properties do you consider most critical for the future use of low-climate and performance-based concrete in practical construction?
8. What needs to be specified or controlled more clearly in order to ensure buildability when using a wider range of concrete mixes and binder compositions?
9. How do these alternative binders change the properties of the concrete? Do the properties improve performance and durability, or do they introduce new risks?
10. When using concrete with a low clinker or high SCM binder usage, what changes (if any) do you recommend for curing and early-age moisture protection?
11. Which alternative binders appear most promising as partial replacements for ordinary Portland cement, especially from a durability and moisture-performance perspective?
12. Could you briefly explain the main moisture-related processes during hydration, drying, and hardening, and what practical problems may arise as concrete loses moisture?

13. What are the most common moisture-related problems observed in concrete structures during construction or early service life?
14. Are there specific construction practices that are particularly critical to prevent moisture-related durability problems?
15. What uncertainties remain in predicting long-term moisture behavior of concrete when using new low-climate binders?
16. Are there any problems with casting, hardening, etc. with concrete with high levels of pozzolans?
17. Do you think current testing methods adequately represent the moisture conditions that concrete experiences in real structures?

A.5 Industry experts

1. What was the first intention of the ERC concept?
2. What is its purpose?
3. Why did the implementation of ERC not work as a standard in the European countries, but now each country needs to decide how to implement it?
4. How will Sweden implement the ERC?
5. Does Sweden currently have the capacity to implement ERC? What capacities are already sufficient and what is lacking?
6. What are the benefits the industry will gain from using ERC? How is this method of designing concrete structures better than the prescriptive-based one?
7. Which exposure classes would benefit most from ERC and in which ones would require more careful concrete mix design consideration?
8. What are the potential risks that should be taken into account regarding this topic?
9. How would the relationship between the contractors and concrete suppliers be affected?
10. Will there be any shifts regarding responsibilities in the design process of a project? Who will be responsible for choosing the ERC classes?
11. When new cements, binders, or concrete compositions are introduced, how should the industry deal with the lack of long-term experience regarding durability?
12. Which types of projects could benefit the most from ERC?
13. What is the Concrete Report and what is its purpose? What is included in the Concrete report?
14. When is the Report Being published? What are the main dependencies or unresolved topics that could delay it?

15. What are the barriers the committee is facing right now? How are they planned to be solved?
16. How do you view the role of standardized testing methods for carbonation and chloride resistance in making ERC usable?
17. What do you see as the main uncertainties or risks when measured test results are translated into ERC classes through calculation models?
18. How do you view the challenge of relating test results on young concrete to durability performance after 50 years?
19. Do you think the complexity of chloride-related assessment, including time-dependent behavior and binder-related aging effects, is a barrier to ERC implementation in practice?
20. Who do you think should carry the main responsibility for testing, interpretation, documentation, and acceptance of ERC-based concrete solutions?
21. To what extent do different binder types create uncertainty for ERC implementation?
22. How important are fresh properties of the concrete for the success of ERC in projects?
23. When implementing ERC in Sweden, might there be any trade-offs between fresh properties and long-ter durability of the concrete? Example: can a mix that performs well in ERC testing still create practical problems on site because of rheology, compaction, bleeding etc.?
24. What additional site-handling requirements do you think ERC-based concrete would create during casting, and what knowledge or training would site management and casting crews need to ensure that the concrete is placed, compacted, and cured correctly?

A.6 Concrete suppliers

1. How do you expect ERC and climate-driven binder optimization to influence your production costs?
2. How do keys influence your willingness to invest in ERC development, and what signals from them would accelerate your investments?
3. What are the biggest practical barriers you see for a smooth implementation of ERC in your company?
4. Which internal capabilities do you need to build or strengthen to efficiently and safely implement this new methodology?
5. How concerned are you that laboratory-verified ERC performance will not be achieved on site due to workmanship, curing, or deviations in fresh properties, and how should responsibilities be shared between you and the contractor in that case?

6. How do you view the increased requirement for tests for the new mixes in terms of time, lab capacity and cost?
7. Will you develop a standard list of pre-qualified ERC mixes, or do you foresee project-specific mix design and testing becoming more common?
8. What support or clarification would you say is essential from the regulations to feel comfortable declaring the performance of the new products?
9. How do you think ERC will change the dialogue between you as a supplier and contractors/designers when specifying fresh concrete requirements?
10. What type of project information (service life, cover depth, construction methods, climate, curing possibilities) do you need at tender stage to propose suitable ERC mixes with predictable fresh properties?
11. Would ERC favor larger suppliers with stronger laboratory and technical resources, or can smaller producers also compete under such a system?
12. Which ones are the most critical when using concrete mixes with higher use of SCMs?
13. How will pumpability and workability of the concrete be affected with the higher use of SCMs?
14. Which tests will the concrete suppliers rely on to control the fresh behavior and ensure workability?
15. Do you foresee any fresh property that will systematically become more difficult to control when moving towards ERC?
16. Are there fresh properties that you expect will benefit from ERC-driven mix optimization? vHow will the increased percentage of alternative binders in the concrete mix affect early strength development and setting time?
17. How do you plan to balance ERC durability requirements with contractors' needs for early formwork removal and construction speed, especially where alternative binders slow down early strength gain?
18. From your perspective, which early-age risks become more critical with ERC mixes?
19. Which binder systems or admixture strategies are most promising for keeping low-clinker concrete both durable and buildable on site?

Appendix B

Thematic analysis sheet

THEMES	CODES	SUPPORTING DATA (QUOTES)	REFERENCE CODE
Benefits	**Freedom to use low climate impact concrete**	"...we are free to use the concrete we want to, and as long as we fulfill the requirements concerning exposure classes or exposure resistance classes, we are pretty free to use this."	**Contractor 0**
		"You can work with the recipes and so on, to reduce CO2 emission or to get a better, uh, concrete structure for the consumer."	**Contractor 2**
		"So today, I mean, we are using much, much more other things than Portland cement, and we have to manage this. Yes, and it's a necessity for the concrete business."	**Industry Expert 1**
		" I think it's it's a good opportunity for for the concrete industry to to move a little bit a little bit further because as it is from for now we are restricted there are a lot of restriction. So I would say it's it's opportunity for us as a as a industry to open up a little bit and to really benefit about the the specific SCM or whatever you working with."	**Concrete Supplier 2**
	Reduce the climate impact	"The concrete manufacturers will be more free to use alternative binders in a higher rate. So, a transition to exposure resistance classes would mean that in many cases we can have a lower, uh, climate impact on the concrete that that we use in our projects."	**Contractor 0**
		"...if we were allowed to use a wider range of w/c ratio in a specific construction part, that would be beneficial for us, I believe, both in climate aspects and as well as economical aspects"	**Contractor 0**
		"...now we are in the situation where the environment, we have to think much more about the environmental question."	**Industry Expert 1**
		"...but the main thing why is because of getting less, uh, for the environmental thing, that you can have exposure classes with less cement, for example, uh, because if you can have less cement and have other mixtures."	**Concrete Supplier 1**
	Optimization	"...of course what we can do is optimize the structures more."	**Contractor 1**
		"...from production point of view and from the manufacturer point of view to optimize the, the concrete."	**Contractor 1**
		"This new thing is with the exposure resistance class opens up for a whole other way of working with concrete and get the right type of concrete for the right structure"	**Contractor 2**
		"If this binder for example performs very well when it comes to chlorides and you can optimize this more and not use this deemed to satisfy or maybe conservative, yes."	**Industry Expert 2**
		"I, I think, uh, performance testing is good, then you can use the materials in the best way not to use uh more than you need, so that that that's I think an advantage.It can be used in that way, um."	**Industry Expert 2**
		"...with more optimized binder content, we can fulfill the same Resistance class but still reduce the shrinkage because we have less binders and we have both better structure because we have less shrinkage cracks and also better economy."	**Contractor 1**
	Economic benefits	"You will save money because the cement will cost a lot of money and you also reduce the amount of cracks, and cracks cost money to to inject."	**Contractor 1**
		"...if we were allowed to use a wider range of w/c ratio in a specific construction part, that would be beneficial for us, I believe, both in climate aspects and as well as economical aspects."	**Contractor 0**
		"Uh, if a company or a mixing company can show that they deliver a product that is better. Then it's quite soon will shift. And you can get more bank for the buck with the new technology."	**Contractor 2**
		"...and then we can have a higher price also because we sell like a performance, not, not, not just a concrete, the gray mass, we sell something different, a more sophisticated product. That's interesting."	**Industry Expert 0**
		"...it depends on on the price of of the binders and and also who is testing."	**Concrete Supplier 2**
		"Of course, if you have less binder, it will be cheaper. You don't have to buy as much, but then, then also it depends on if you have to test a lot, that might come with a cost also, so."	**Industry Expert 2**
	Projects or exposure classes that benefit more	"Chloride is, is the one that we think and discuss the most because it's the easiest step."	**Contractor 3**
		"One example is the garage, for example."	**Contractor 3**
		"...it would be good in all projects."	**Contractor 0**
		"...that could be like in a garage, for example."	**Contractor 0**
		"...the other ones that I think would benefit is the de icing and salt XD and XS."	**Contractor 1**
		"...in, in the beginning, when this concept is introduced into the standards, it will be, I think, mainly to really large projects."	**Industry Expert 0**
		"I think for the more, more not so harsh environments like in, in buildings and in, in, in, uh, in, uh, there will be, it will go faster because, because it's always like that if you don't have too much environmental, uh, attack on, on it, it, it, it's easier."	**Industry Expert 1**
		"...the easiest one is perhaps in what you call it, parking lots in the parking structures, garages."	**Concrete Supplier 2**
		"I guess to try to answer your question, it's perhaps fair to say that the bigger project is going to beneficial more if if we're thinking that we're going to make a special mix for that."	**Concrete Supplier 2**
		"No, not housing like more where it's exposed to, to uh chlorides and so on. Where you can uh use to the the positive effects on chloride, for example."	**Industry Expert 2**
		"We have been uh, I mean, adopting this sort of the concept a little bit in the past, we have, uh, for instance, for garage, concrete, concrete for garage structures in XD3, we offer to some customers at least pre pre-tested or pre-qualified concrete with higher water-to-cement ratio as compared to the standard with some slag."	**Industry Expert 1**
		"And where, where is the most benefit we can, we can, we can scale it from. So, um, we have to find out, is it, for example, with, example, uh, regarding swimming areas, for example, or is it regarding uh buildings quite close to the, to the sea."	**Industry Expert 1**
		"For example, if we want a concrete that has a good resistance against chlorides, then this ERC concept is really a way to test it"	**Industry Expert 0**
	Reaching environmental goals?	"...when it comes further on like, 2030 and beyond, I do think that in order to use concrete and achieve the goals that we have set up, all concrete has to be a quite high degree of alternative binders."	**Contractor 3**
		"If we look at the roadmap to zero emission, I think this is a deal breaker to reach the goals that you can go from this traditional way with the exposure classes to more see what kind of properties the concrete needs to have."	**Contractor 2**
		"That could, that could be one and and then also, I mean that also connected with the more greener achievement that may be as well."	**Concrete Supplier 2**

	Introduce new binders more easily	"That, that's actually one of the starting points for this, uh, ERC concept in the, in Europe also that it should, uh, make it easier to, uh, introduce new binders."	**Industry Expert 0**
		"...it will be easier, it will be easier to introduce new materials, uh, because you can make tests, uh, to, to see what it is."	**Industry Expert 1**
		"...it could be, uh, a way, an easier way to introduce a new type of materials that we don't have, uh, as much experience of."	**Industry Expert 2**
		"...it's good that it will definitely give us the opportunity to, uh, to introduce maybe new binder combinations a little bit quicker and also to uh to uh by, by testing to offer customers better solutions maybe yeah."	**Concrete Supplier 3**
		"...it sounds good, uh, that you can, uh, easier use new binders."	**Concrete Supplier 0**
		"...of course what we can do is optimize the structures more and be able to, for instance, use other materials than only cement or other new materials that will come in the future."	**Contractor 1**
Barriers, risks, uncertainties	**More complexity**	"...we will have a greater wide of a greater range of products at our construction sites that are performing differently from each other."	**Contractor 0**
		"...it's always quite hard in the beginning to come with a new technology or a new material because you have to have a, a certain demand to get the prices down."	**Contractor 2**
		"So it, it gets complicated, I think. Already now it's complicated."	**Industry Expert 0**
		"...it's making it quite complex because and the the knowledge about it as to be there as well."	**Concrete Supplier 2**
		The diversity of the product that we will be using in the future will be a challenge for us"	**Contractor 0**
	Struggle of the smaller companies to comply	"...we will be able to implement this, but not all small contractors that still have to fulfill the regulations."	**Contractor 0**
		"...smaller producers, I think that they are, they will have problems with this concept because it's something completely new for them."	**Industry Expert 0**
		"...every concrete producer cannot, cannot, uh, test all these performances, uh, regularly."	**Industry Expert 1**
		"More beneficial to be a large player than a small one in this case."	**Concrete Supplier 2**
		"...but this testing will not be so, so much if it's very, uh, high demands on testing. And this will also mean that only the big companies will be able to do it, because it comes with knowledge and a cost, so."	**Industry Expert 2**
		"...it will be probably difficult to find a laboratory laboratories to test. To perform the testing, you know."	**Concrete Supplier 3**
		"...if we need to test. I, I think that, I mean, it would be easier for bigger companies to handle. It would be a bit more advanced probably."	**Concrete Supplier 3**
		"There are a cost for all the testing also when you don't have your own laboratory."	**Concrete Supplier 0**
	Long transition time	"...in the moment how we're not going to use because it will be a very long transition time."	**Contractor 1**
		" So I think within 10 years perhaps."	**Industry Expert 0**
		"That way things are not going to happen overnight."	**Concrete Supplier 2**
	Lack of knowledge	"I think this is very new to everybody. And the designers are the last ones on the train."	**Contractor 1**
		"...the designer in Sweden have never been good in material design, never, neither in ERC or in exposure classes."	**Contractor 1**
		"If we have the people on site that are not engaged in this change, it's an increased risk that the casting will be faulty, that we will have more errors."	**Contractor 3**
		"...we don't have any structures that we actually know that it will perform as it does because we have no reference structures to look at."	**Contractor 2**
		"...the structural engineers, they are. Uh, They're quite slow to adapt new things."	**Industry Expert 0**
		"Also in our, our existing system, we have this problem, and, and here the, if you have small amounts of smaller amounts of, of uh of um of uh cementitious materials, other things, then there is normally not a problem, but if you go to higher amounts and we are with the environment in question, we are, we are going towards higher amounts, then this will be more and more accentuated this lack of, of um education on on the uh in in the business."	**Industry Expert 1**
		"...those who have knowledge about this, like, uh, Skanska and Thomas Betong and that kind of thing which have a very good knowledge on the material side too, uh, but if you go to a small, small, uh, concrete producer in here in in *Inaudible* where I live, where I live, well, they have heard we can, we can make a very good concrete here with a lot of slag and, and that, but we didn't know about this fresh concrete, fresh properties or, or, or that kind of thing."	**Industry Expert 1**
		"The prescribers are are just doing looking at the whole thing and just doing the same same thing all over again. That's at least how it used to be."	**Concrete Supplier 2**
		"I mean, of course there are going to be prescribers that actually have this capability and a lot of knowledge, but I mean looking at the overall, I would say that they have a lack of knowledge."	**Concrete Supplier 2**
		"Uh, I think also the, the the knowledge about concrete is decreasing in the projects. They are often looking at us as the experts. They, they don't think they have the the knowledge"	**Concrete Supplier 0**
		"The designer in Sweden have never been good in material design, never, neither in ERC or in exposure classes."	**Contractor 1**
		"I wasn't that into it before, uh, you mentioned it for me, and then I, but then I have read about it"	**Concrete Supplier 1**
		Are you familiar with the exposure resistance classes? "Yes, not in detail, I should say, not in detail. I, I do know about the intention, uh, but I don't know the detail exactly how it works."	**Contractor 3**
		"I can't say I'm overwhelming with knowledge about it, but I get I I know and I haven't read Eurocode and know exactly how it's going to be, but in in I have a little knowledge about it and yeah."	**Concrete Supplier 2**
		"I have tried to look, to look up, uh, some information, uh, about it, so I've tried to. To, to learn a bit now shortly"	**Concrete Supplier 0**
		" We haven't thought a lot about the ERC concept so far actually, so it's, uh, it was good for me to think a bit about it before today's meeting."	**Concrete Supplier 3**
	Skepticism of the clients	"...we have some clients that say that oh we don't need, we don't want any SCMs, uh, because we, we don't believe in it."	**Contractor 3**
		"...some clients, they want to know that a new technology works before they want to try it in their own project."	**Contractor 2**
		"I heard this argument from some of the clients, so they were very hesitated to try the new concretes because, they were more comfortable with the old ones that they knew worked."	**Contractor 2**
		"Uh, we have a conservative business. Everything new isn't uh taking the yahoo, uh. People when they're used to working with one type of material, they are quite suspicious when you come with a new type."	**Concrete Supplier 0**
Requirements for implementation	**Increased need for knowledge and dialog**	"I think there will be a need for an increased dialogue between us contractors and the concrete manufacturers in order to receive the right products products on our construction sites"	**Contractor 0**

		"...it's very important that the entrepreneur have a good relation with the supplier and the dialogue and and that we require the adequate amount of information before we do changes, especially when it comes to more critical structures."	**Contractor 3**
		" I think it depends on the contractor actually. If it's a big contractor that has a technical staff behind, then it's easier to have a good communication."	**Industry Expert 0**
		"Even more possibilities to develop the customer to fulfill the customer wishes, I would say, so we have more discussion about what kind of properties that we will require for different situations."	**Concrete Supplier 3**
		"...there will be a demand for further educational within the company."	**Contractor 0**
		"...there will be more responsibilities and more knowledge needed from the structural designers to know more about the concrete that they are going to use"	**Contractor 1**
		"...we need more knowledge and people that have knowledge to in order to to who can ask the right questions in this change."	**Contractor 3**
		"More knowledge is needed."	**Contractor 2**
		"...we have to have a lot of education and to really teach, especially those who prescribe. How to prescribe according to this system."	**Industry Expert 0**
		"I think there will come some kinds of, uh, there will be needed some kind of, guidance on this for the structure engineered..."	**Industry Expert 1**
		"OK, they have, there, there will be some changes, so they have to, they have to learn to, but not not extremely different."	**Industry Expert 1**
		"So my thoughts about that, it's that the prescribers are a little bit more up to date and and looking at the specific construction trying to lower the CO2 level."	**Concrete Supplier 2**
		"...the knowledge about it as to be there as well."	**Concrete Supplier 2**
		"And that makes it for us also, we have to know much more now and uh. Our personnel also have to be educated so they uh really know."	**Concrete Supplier 0**
		"Uh, and we have the, uh, when we see what the regulations are, we know what kind of equipment we need. And then we, we, we know how, what we have to have to do to, to, to test different recipes and what we have to do and we can't start with that until we know the regulations of course we can start, but then we just have to, to guess."	**Concrete Supplier 1**
	Need for tests	"...much more tests will be done, needed to be done because all the binders cost much more and will be much more expensive in the future."	**Contractor 1**
		"...with all this kind of new binders you will have, you need to prove it worthy to use in concrete."	**Contractor 1**
		"...when we introduce new new type of SCMs, a new type of cements, we need a system to ensure that we have the right properties of the new combination."	**Contractor 3**
		"...you have to show tests and so on, so that can make it a little bit more expensive in the beginning and also testing to validate that the new structure will have the all, all of the performance that the client wants."	**Contractor 2**
		"...you have to test different or new mixes to see if you can work with the concrete."	**Contractor 2**
		" ...if we change, it could be the aggregates, for example, or the admixtures or the binders or so on, then we have to make a sort of pre-testing to both get the fresh properties of the concrete and the hardened properties of the concrete correct so that we can fit them into the system."	**Industry Expert 0**
		"...if you can show that if it's enough that the cement company, for instance, qualify a certain binder combination like they do today with they can do it with the EPCC concept. If you do it the same way, then it will not be a lot of testing."	**Concrete Supplier 3**
		"So it, it, it will require time and advanced lab equipment, for example, uh, so it would be higher initial costs."	**Concrete Supplier 1**
	Capacities of the industry	"...even now it's possible to use function testing in order to do something similar to ERC. There are possibilities within the standards, but It's not obvious how it's done and and how to to to to do it properly in projects, which means that we do, we do put a lot of hope in the ERC that when it comes it will be easier, but the question is also then how is it implemented."	**Contractor 3**
		"...not always, but often a, a big client that has some knowledge in new areas so that they have an organization that can also look into the future and what's coming because smaller clients don't have the capacity."	**Contractor 2**
		"In Sweden, we have done a very a lot of field studies on the west coast. Uh, but, but if you look, if you look at the the cover tables in, in, in the regulations, uh, they don't give any, any, cover for the west coast. They give for the east coast on which there has not been done very much work, so it's a little bit contradictory."	**Industry Expert 1**
		"I mean if you're if you're a bigger supplier with lot of capability and and and are able to do this kind of testing."	**Concrete Supplier 2**
		"if you want to develop the concrete you have to, It's it's good to have the capability yourself."	**Concrete Supplier 2**
		"But, but we, we only have small laboratories to, to, to more production-based testing. Some, some testing, but, but that's only uh to uh pressure test, such some tests."	**Concrete Supplier 0**
		"There are many labs also the suppliers, um, to the *Inaudible* producers like *Inaudible* they also have their own labs doing their tests. You have also the cement suppliers doing their own tests, so I think there are a lot of capacity for that, absolutely."	**Contractor 1**
Changes in the industry	**Shift in responsibilities**	"The responsibility for fulfilling the exposure resistance classes means more responsibility on the manufacturers of the concrete"	**Contractor 0**
		"...structural engineers will still have a, a great impact on prescribing the concrete"	**Contractor 0**
		Should the structural engineer be the one responsible for prescribing ERC? "I think so. I think so. That's the way it has been done in Sweden"	**Industry Expert 0**
		"I think it, it will be better because the designers, they should only design with regard to what they need, and this is to give them a better better tool to say what they need and then it's up to the materials to fulfill this by making a concrete that has the right durability or something like that, yeah."	**Industry Expert 1**
		" ...if you look at the at the, the designers, it will, it will make the design firms, it will make their job much easier because they only have to decide what cover do I need and what kind of, uh exposure resistance class."	**Industry Expert 1**
		"...so perhaps we're looking at a new type of prescribers, prescribers who also have knowledge about this."	**Concrete Supplier 2**
		So you'd say that there's an increase in responsibility when it comes to the concrete producer side? " Yeah, I think so because, uh, our customers, they don't have the education in concrete."	**Concrete Supplier 0**
	Who is liable if something goes wrong?	"...it would be harder to say who is the actual responsible, who has the actual responsibility."	**Contractor 0**
		"It depends on the contract."	**Contractor 2**
		" I mean if he say, OK, we have done everything, it's a perfect concrete, it fulfills everything and we have done all the tests, and anyhow we get, we get one problem or another, then it's a contract business between the, the, them because."	**Industry Expert 1**
		"If if there's been a failure, that it needs to be investigated where the failure is, uh, but I guess if, if, if, uh. The concrete fulfills all the requirements. It's, it's hard to to. Put their own responsibility on the concrete producer."	**Industry Expert 2**
	Competitivness	"There will be more competitivness because you have a bigger freedom to change the recipes."	**Contractor 2**

Status of development and regulatory progress	**Intention with ERC**	"...perhaps in areas where you have a lot of populations or where you are going to build a lot, perhaps you can see some new suppliers coming up."	**Contractor 2**
		" ...it will be, uh, then it's a major competitive advantage for us, because if, if, if we can, we can find out this, uh, it's a business impact that we have something that no one else has."	**Concrete Supplier 1**
		"...if we can deliver tender with low cost and low carbon impact, that would be great beneficial for us in the market. Because more and more customers are interested in having buildings built with lower carbon impacts. So, it's more often uh a competitive thing."	**Contractor 0**
		"...intention was that there should be an EN 206 that should describe how the concrete should be composed, but how, how to test concretes with different test methods to with the new mixes."	**Industry Expert 0**
		"...the meaning was that with this system, all these national provisions, they should be removed."	**Industry Expert 0**
		Would you say that also ERC firstly was intended to be some sort of harmonization across Europe? "Yes, yes, yes. It was thought that this should be because, uh, today, I mean, you're looking at there is the durability rules are quite different in, in, in, in all countries."	**Industry Expert 1**
		"I think it um was to, to sort of have a harmonized durability requirements, to have the similar requirements throughout Europe, uh, and base it on, on performance, in that case."	**Industry Expert 2**
		"So now, I think that is our responsibility as durability experts to see that this is not done in a way that the, the environmental question cannot overrule the durability question because they are very closely linked together."	**Industry Expert 1**
		"The intention was that the producer of the binder, they should test their binder."	**Industry Expert 0**
		"ERC is mainly a structural problem with the concrete covers and then it's the concrete composition. So this execution part is not really covered in the ERC."	**Industry Expert 0**
		"...the ERC and this er this er er concept, it can be used for several things. One is to use it purely, I mean, to, to, to test concrete instead of saying that it should contain this and this and have this water to binder ratio or something like that. But the other one is that we are using this concept to uh, we are adapting, uh, there is always the possibility to take, to create, deemed to satisfy rules."	**Industry Expert 1**
	Barriers with regulations	"...all countries in Europe have chosen. There are quite large differences, especially in the structures exposed to chlorides..."	**Industry Expert 0**
		"...there are models that are used to predict, for example, chloral ingress. Uh, and in, in this group they're not agreeing on how the model should be defined."	**Industry Expert 0**
		"Ryle group and this FIB group they have some. Have made some comparisons between different models, and they see that this model, it will underestimate the chloride ingress. And that is not really good because then we overestimate the durability, we overestimate the service life."	**Industry Expert 0**
		"...the national specialists in concrete, they want to have their own rules and they want to be like the author of the rules."	**Industry Expert 0**
		"...when you come to especially durability experts, there are a lot of, there are uh sometimes too many experts and everybody has his own model and his own thinking and his own thing, so, so. They are, er, some of them are very strong-minded"	**Industry Expert 1**
		"...it was, uh, hard to, to, uh, agree on, on, uh, all the details in the concept. I don't know exactly what what it was, but it can be from, from models and then test methods and all, all this also with, um. Verification of the, of the, the durability, so."	**Industry Expert 2**
	Concrete report	"Yes, you know, we are, we are working on, on, on, uh, in this project with the Betongföreningen, uh, where we are, uh, we are, uh, discussing this, it's, uh, both, both sides, both the design side and the, the material side. And this group will come up with a proposal on exactly how they think this should be implemented."	**Industry Expert 1**
		"...it will be like a description of the concept from the beginning. In the first chapter, and then we will talk a little about these different models to make predictions. Then we will talk about how to classify different concrete based on testing. We will have a part that is about how to prescribe concrete according to this exposures resistance class system. There will be a little about how the contractor should work. And then of course it could be a future research."	**Industry Expert 0**
		"...our intention is that it should be published this year. Hopefully at the end of 2026."	**Industry Expert 0**
		"...we will be very clear that this is like a, a first version of the report and that it should be revised when we have better knowledge."	**Industry Expert 0**
		"Well, I think the, the report should be published at the end of this year."	**Industry Expert 1**
		"Yeah, it's good, yeah, kind of a description of the concept and a guideline for how to, to, uh, test concrete according to, to the , for example, resistance class for carbonation. And maybe give some guidance on which type of materials that can be used."	**Industry Expert 2**
		"Just because it could not be harmonized. So this, and, and I think that what I heard was that in the, in the version that will be launched of the Swedish standard SS 137003, it will not incorporate the ERC uh concept now."	**Concrete Supplier 3**
	Future	"...the results from this concrete report should be in some way implemented in the SS 137003."	**Industry Expert 0**
		"I mean, if, if several countries are deciding on how to do it, in the end, perhaps in 5 years, 10 years, you can look at how, how have other, other countries done this, and that will be the start for a harmonized part of it."	**Industry Expert 1**
		"Like the, the first step is like we're doing now, uh, to, to create these guidelines and how we can apply it. And and one part of that is to to have these tables with the cover thickness, uh, for the different resistance classes, I guess that one will go into like the natural application of the Eurocode. So that, that's, uh, is one step. hen, uh, we need to, to, uh, introduce it in our, uh, National application document for concrete in 206."	**Industry Expert 2**
		"I think when it will be introduced, it, if it's not just an appendix in the standard that this is a possibility, uh."	**Industry Expert 2**
		"...it will not be mandatory to use this exposure resistance class system. We will still have this old system with the prescribed rules."	**Industry Expert 0**
Properties of the concrete affected	**Workability**	"So it have to be a similar workability, I think. To, to get a smooth transition towards climate optimized concrete, you, you, you can't have too, too much difference."	**Concrete Supplier 0**
		"I have also heard a couple of complaints regarding the workability, stickiness in some type of binders or some kind of concrete."	**Contractor 3**
		"It depends on the the properties on the SCMS that you're putting in. I mean it's a, it's a, it's a different kind of binder or so for sure it's going to be, it's going to differ."	**Concrete Supplier 2**
		" Well, of course, the workability, for example, the slump or how to measure the workability, that's really important."	**Industry Expert 0**
		" Yeah, both good and bad, I could say, yeah, in the beginning when we started and we took just 10% with a different binder. Uh, they, they didn't, uh, they didn't actually felt something. It was just like the ordinary, but as we go further and further, like 20% and 30% and stuff like that, things happen, and they notice that the, the product is not the same."	**Concrete Supplier 1**
	Increase the w/c ratio	"...instead of 380 kg of cement per cubic we use 300 going down and go up in what w/c ratio but still have the same resistance class as the previous concrete."	**Contractor 1**
		"...for parking decks, both Heidelberg and Thomas Betong has shown that with a water/cement ratio of 0.5 together with GGBS, we can reach the same chloride resistance as with only cement and water/cement ratio is 0.4."	**Contractor 1**

		"...if you think about the ERC concept we, I would like to see that we go towards higher water/cement ratios, which also means that we get a more workable mix usually."	**Concrete Supplier 3**
	Strength development	"Yeah, all this supplementary cementation materials slow down the, the strength development, yeah. Sure. Uh, most of them, at least that we use today, so."	**Industry Expert 2**
		"...as is seen already today, today, it's a problem with the early, early strength is one permit, which is, it's critical."	**Concrete Supplier 3**
		"There are really with all kinds of SCMs, there are different kinds of beneficial and more negative stuff. And I mean short term strength could be one, but it could also be beneficial. I mean in some cases don't want to develop as much."	**Concrete Supplier 2**
		"I don't think so now with with slag, when we use slag. Then you almost have, a, a high, at least in the summer, uh. When, when you come into winter, the short, the short term strengths is, is slower, uh."	**Concrete Supplier 0**
	Long-term strength	" If you, if you think about these SCMs, often they have uh, higher long-term strength. Uh and then uh and then, compared to Portland cement."	**Industry Expert 2**
		" I mean, the strength is usually not a big issue."	**Concrete Supplier 3**
		"Uh, my view is that you get uh a, a higher hard strength. When when you go with slag when you go past 28 days, you still have an increase."	**Concrete Supplier 0**